
STEERING THE HERD: A FRAMEWORK FOR LLM-BASED CONTROL OF SOCIAL LEARNING

Anonymous authors
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ABSTRACT

Algorithms now operate as powerful information mediators, from social media feeds and targeted advertising to the rapidly expanding use of LLMs. Their mediation creates a coupled decision process: agents act on private, algorithmically mediated signals while simultaneously learning from the visible choices of peers. We introduce a model of controlled sequential social learning in which an information-mediating planner (e.g., an LLM) controls agents’ information structures while agents continue to learn from earlier decisions. The planner is either altruistic, seeking to improve social welfare, or biased, seeking to induce a preferred action. This framework defines a new optimization problem for social learning that combines dynamic programming, decentralized action choices, and Bayesian belief updates. We prove convexity of the altruistic value function and characterize optimal policies for both altruistic and biased planners, identifying the precise tradeoffs between control costs and payoffs from induced choices. The characterization shows that optimal mediation has sharply distinct regimes: the planner may invest the maximum allowed resource, invest nothing, increase investment with belief, or decrease investment with belief. In key belief regions, the biased planner optimally obfuscates agents’ signals. Even under stringent transparency constraints—information parity with individuals, no lying or cherry-picking, and full observability—information mediation substantially shifts social welfare in either direction. We complement the theory with simulations in which LLMs act as both planner and agents. The LLM planner exhibits emergent strategic steering of public opinion that broadly matches the predicted policy structure, while systematic deviations reveal the role of non-Bayesian reasoning consistent with cognitive patterns in humans and LLMs trained on human-like data. Together, these results establish controlled social learning as a tractable and behaviorally relevant framework for analyzing and regulating LLM information mediators.

1 INTRODUCTION

Large Language Models (LLMs) are becoming central information mediators in socially critical domains, from search engines and news aggregators (Zhang et al., 2024a; Gao et al., 2024) to personal assistants offering medical or political advice (Huo et al., 2025; Haupt & Marks, 2023; Schiele et al., 2024; Argyle et al., 2023). This role matters because LLMs already display persuasive (Potter et al., 2024; Carrasco-Farre, 2024; Rogiers et al., 2024), manipulative (Williams et al., 2025; Liu et al., 2025; Jones & Bergen, 2024), and strategic (Lorè & Heydari, 2024; Payne & Alloui-Cros, 2025; Zhang et al., 2024b) behavior in user interactions. At scale, such influence is never isolated: it enters social systems in which people observe, imitate, and learn from one another. The resulting feedback between algorithmic mediation and peer observation can decisively amplify or counteract a model’s effect.

A simple recommendation setting makes the mechanism concrete. An LLM-powered system recommends a new restaurant to a sequence of potential customers. The restaurant’s quality—the true state of the world—is either good or bad, but neither users nor the system directly observe it. Each user receives a personalized recommendation, which functions as a private signal, and then decides whether to patronize the restaurant. The recommendation system (the *planner*) can spend resources to make this signal more informative, for instance by generating a highly tailored review from the

user’s known preferences. This investment captures a real control cost: market research, user surveys, or data acquisition needed to tailor information effectively.

The user’s decision, however, is not determined by the private signal alone. Users also observe predecessors’ actions—if the last ten users patronized the restaurant, the next user rationally infers that they likely received favorable information. This *social learning* channel strongly shapes opinion and behavior in settings ranging from vaccine adoption (e.g., Rao et al. (2007); Bauch & Bhattacharyya (2012)) and technology adoption (e.g., Gillingham & Bollinger (2021); Weber (2012)) to political and moral judgment (e.g., Brady et al. (2021); Guilbeault et al. (2018)). The central contribution of this paper is to characterize the strategic interaction between a planner that controls private information and agents who learn socially. The planner must account not only for the current user’s response, but also for the information *externality* that each induced action creates for all future users.

We distill this interaction into a tractable formal model of *controlled social learning*: an algorithmic planner strategically chooses, at a cost, the precision of private signals sent to a sequence of agents. The planner’s influence is deliberately constrained and therefore especially revealing. It does not falsify information; it chooses how clear and informative each signal should be. This model lets us answer three core questions directly: how a planner can steer collective beliefs, how optimal strategy changes with public opinion, and how such mediation ultimately affects social welfare.

The answers turn decisively on the planner’s objective. An *altruistic* planner, such as an ideal recommendation system, maximizes user utility by helping users choose correctly: patronize the restaurant if and only if it is good. A *biased* planner, such as a system rewarded by the restaurant, instead seeks to induce a specific action—patronage—regardless of the true state.

This distinction is related to, but sharper than, the usual language of alignment. The altruistic planner is aligned by construction with the agent’s *expected utility* under uncertainty. The biased planner is not, because its objective is state-independent; nevertheless, its influence need not always harm realized outcomes. When the true state favors the planner’s preferred action (e.g., the restaurant is in fact good), the biased planner’s intervention aligns with the agent’s *realized utility* in that instance. Our analysis covers this full range, treating altruistic and biased planners under both realized alignment and realized misalignment with users.

Contributions and Outline This paper delivers the following principal contributions:

1. **A New Theoretical Framework for Controlled Social Learning.** We introduce the first formal model that integrates a dynamic control problem for a centralized information planner with sequential social learning. The planner strategically chooses the precision of agents’ private signals to optimize an objective that is either altruistic (maximizing social welfare) or biased (inducing a specific action).
2. **A Complete Characterization of Optimal Planner Policies.** We characterize optimal policies for both altruistic and biased planners as functions of the evolving public belief. In the altruistic case, the characterization rests on a new proof of value-function convexity. These results identify the strategic tradeoffs a planner must make when each controlled signal creates informational externalities for future agents.
3. **Empirical Validation and Strategic Analysis Using LLMs.** We run simulations in which LLMs serve as both planner and agents. The experiments establish three findings: (a) planners that account for social learning can shift public opinion and social welfare far more than myopic planners; (b) the LLM planner’s emergent strategy closely tracks our theoretically optimal policy structure despite non-Bayesian LLM agents; and (c) LLMs exhibit strategic reasoning sophisticated enough to exploit social learning, underscoring both their promise and their societal risk.

Section 2 positions the contribution relative to prior work. Section 3 presents the formal model, Sections 4 and 5 derive the optimal policies, Section 6 validates and stress-tests the theory with LLM-based simulations, and Section 7 concludes. Proofs and additional results appear in the appendices.

2 RELATED WORK

Our work connects four literatures—social learning, information design, online persuasion and RL, and LLMs—and occupies a distinct intersection among them.

Social Learning Bikhchandani et al. (1992) and Banerjee (1992) introduced the canonical sequential social learning framework in economics, and subsequent work has characterized these dynamics in more general environments (e.g., Smith & Sørensen (2000); Arieli & Mueller-Frank (2021)). This literature also analyzes alternative forms of social information and systematic biases or misperceptions among agents (see Dasaratha & He (2022); Bistriz et al. (2022); Eyster & Rabin (2010) as well as Bikhchandani et al. (2024) and the references therein).

By contrast, the *control* of social learning remains underdeveloped (Wei & Anastasopoulos, 2022; Smith et al., 2021; Krishnamurthy, 2012; Bhatt & Krishnamurthy, 2021). The closest works are Wei & Anastasopoulos (2022) and Smith et al. (2021). Wei & Anastasopoulos (2022) assumes two-way communication, allowing agents to report private signals to the planner in exchange for action recommendations. Smith et al. (2021) studies a benevolent planner who directly alters agents’ choice rules and derives a decentralized pivot mechanism to improve learning. Our model removes these strong forms of intervention: it requires no two-way communication, preserves agents’ control over their actions, and therefore directly targets information-mediating algorithms that are often invisible or black-boxed to users.

Information Design Our planner optimizes by choosing the precision of agents’ private signals, making the model a constrained instance of information design (Bergemann & Morris, 2019). This literature rose to prominence with Kamenica & Gentzkow (2011)’s model of Bayesian persuasion, where a planner selects the signal structure observed by a rational Bayesian agent facing an unknown state. In our setting, the planner solves a costly and constrained information-design problem in every period: it chooses a binary symmetric signal at a cost. The decisive difference is that receivers also learn socially, so each signal affects not only the current receiver but also the belief process faced by future receivers.

Only a small set of papers studies information design within sequential social learning (Arieli et al., 2022; Wu et al., 2025). These works choose an information structure once at the onset and then hold it fixed across the agent sequence. Arieli et al. (2022) analyzes a self-interested sender choosing a general Bayesian-persuasion signal structure, while Wu et al. (2025) restrict the structure to be Blackwell more informative and maximize the probability of a correct cascade. We instead solve a genuinely dynamic problem in which the sender chooses a new structure for each agent, creating a richer and strategically sharper class of policies.

Online Persuasion and RL Recent work at the interface of online learning and Bayesian persuasion/information design studies online sequential persuasion with unknown environment parameters. In these models, a sender repeatedly interacts with short-lived receivers without knowing the prior over states and/or receivers’ utility functions (e.g. Castiglioni et al. (2020); Bacchiocchi et al. (2024); Castiglioni et al. (2021); Agrawal et al. (2023)). A related literature analyzes sequential persuasion in dynamic environments or MDPs, adding state transitions and/or receiver foresight; these models emphasize history-dependent or promise-based signaling policies and often connect to reinforcement learning (e.g. Wu et al. (2022); Su et al. (2022); Bernasconi et al. (2023)). Their central objective is learning-based performance—sublinear regret relative to the best signaling scheme in hindsight and/or approximation guarantees under computational constraints.

Our model isolates a different source of sequential dependence. The planner does not learn unknown parameters; dependence arises because receivers learn from one another. The planner also has no informational advantage: it observes neither the underlying state nor private signals. Under these restrictions, we nevertheless characterize optimal planner policies. Thus, although the model is sequential, it is not an online-learning or RL environment: actions do not change the payoff-relevant state, there are no reward externalities, and the planner does not explore unknown environmental parameters. The planner updates only through agents’ observed actions, which encode social learning and belief propagation. Sequential dependence therefore comes entirely from information externalities.

LLMs A growing literature studies LLMs in information design (Harris et al., 2025; Cheng & You, 2025; Raifer et al., 2022; Bai et al., 2024; Duetting et al., 2025). These papers focus on one-on-one

interactions and therefore leave the social-learning channel unmodeled. The closest work is Duetting et al. (2025), which studies information design where a sender chooses both a signal structure and a 'framing' that shapes the posterior belief of a possibly non-Bayesian receiver. Their LLM-based oracle addresses the computational difficulty of optimizing over framings and builds on evidence that LLMs can mimic humans as economic agents (Dillion et al., 2023; Horton, 2023). Our paper extends this agenda to repeated interaction with social learning; we adapt parts of their methodology for the empirical study in Section 6.

3 SOCIAL LEARNING MODEL

We model a planner interacting with a countable sequence of Bayes-rational agents indexed by $i \in \mathbb{N}_{>0}$. At time $i = 0$, nature draws a fixed, *unknown* exogenous state $\omega \in \Omega := \{G, B\}$, with $\mathbb{P}(\omega = G) = b_1$ known to all participants. In the recommendation example of Section 1, G is precisely the event that the restaurant is good.

At each time i , the planner chooses how much to invest in personalization for agent i and then provides a private signal $s_i \in \Omega$ to support the agent's decision. The signal is binary and matches the state ω with probability $q_i \in [0.5, 1]$, i.e. $\mathbb{P}(s_i = \omega) = q_i$. We call q_i agent i 's signal precision.

Conditional on ω , each signal is independent of other agents' signals and of the history. Agent i then combines this signal with the observed actions of predecessors $j < i$ and selects an action $a_i \in \Omega$. She receives utility 0 when $a_i = \omega$ and utility $-C$ otherwise, with $C > 0$, and therefore chooses the action that maximizes her expected utility. In the running example, this means patronizing the business exactly when the state is G .

3.1 AGENTS' DECISION PROBLEMS

Each agent observes every predecessor's action and signal precision. We denote this history by $\mathcal{H}_i := (b_1, (q_j, a_j)_{j < i})$. Because the planner and all agents share $\mathcal{H}_i$, they also share a *public* belief about ω that updates after each action. Immediately before agent i acts, this belief is $b_i = \mathbb{P}(\omega = G | \mathcal{H}_i)$; before any action, it equals the prior b_1 . As in Banerjee (1992) and Bikhchandani et al. (1992), $(b_i)_{i \in \mathbb{N}}$ is a Markov process and a sufficient statistic for the full history $\mathcal{H}_i$ (see Appendix B.3 for proof).

Given $\mathcal{H}_i$ (equivalently, by Markovianity, b_i) and q_i , agent i chooses $a_i \in \Omega$ to maximize utility. If the two actions tie, she follows her private signal s_i . Combining the public history with signal s_i of precision q_i , agent i forms the *private* belief $\tilde{b}_i$ (derived in Appendix B.1):

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q_i}{1+2b_iq_i-b_iq_i} b_i & s_i = G \\ \frac{1-q_i}{b_i+q_i-2b_iq_i} b_i & s_i = B \end{cases} \quad (1)$$

The agent then chooses the action corresponding to the state that is more likely under posterior belief $\tilde{b}_i$: $a_i = G$ if $\tilde{b}_i > 0.5$, $a_i = B$ if $\tilde{b}_i < 0.5$, and $a_i = s_i$ if $\tilde{b}_i = 0.5$. Substituting for $\tilde{b}_i$ yields the sharp action rule:

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases} \quad (2)$$

Subsequent agents observe a_i and incorporate it into the public belief according to the update below (see Equation (14), Appendix B.2):

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (3)$$

Remark 1. When $1 - q_i \leq b_i \leq q_i$, Equation (2) implies that agent i 's action perfectly reveals her private signal, so the updated public belief equals agent i 's private belief. Outside this interval, the private signal does not affect the action, and the public belief remains fixed. The process then enters the absorbing state known as an information cascade or herding in Bikhchandani et al. (1992); Banerjee (1992). At such points, society stops learning from agents' private information.

This decision rule gives agent i the expected utility $-C\mathbb{P}(a_i \neq \omega|b_i, q_i)$, which reduces to the following compact form (see Appendix C.1):

$$-C\mathbb{P}(a_i \neq \omega|b_i, q_i) = -C \min(b_i, 1 - b_i, 1 - q_i). \quad (4)$$

3.2 THE PLANNER'S PROBLEM

We analyze two planner objectives. An altruistic planner seeks to induce correct actions ($a_i = \omega$), while a biased planner seeks to induce a fixed action, taken to be G , regardless of ω . We use subscripts A and B for these planners. In both cases, the planner controls each agent's private-signal precision. The non-negative, increasing, continuous, and concave function $\beta(\cdot)$ gives the cost of the chosen precision, and the planner's information set is exactly the common history $\mathcal{H}_i$.

Altruistic Planner At time i , the altruistic planner chooses precision q_i for agent i and pays $\beta(q_i)$, with $\beta(p) = 0$ for baseline precision $p \in [0.5, 1)$. Because decreasing precision never benefits the altruistic planner (Appendix C.9), we can focus without loss on additional investments above p . The added cost rises with precision but has decreasing marginal cost. Agents know both p and $\beta(\cdot)$.

The altruistic objective is social welfare net of precision-investment costs, where social welfare is the expected total utility of the agents. Let $r_A(b_i, q_i)$ denote the instantaneous reward from public belief b_i and signal precision q_i for agent i . Starting from b_1 and following policies $\pi = (\pi_i)_{i=1}^\infty$ with $\pi_i(\mathcal{H}_i) = q_i$, the planner obtains the expected total discounted utility below for discount factor $\delta \in [0, 1)$:

$$V_A^\pi(b_1) = \sum_{i=1}^{\infty} \delta^{i-1} r_A(b_i, \pi(b_i)), \quad \text{where} \quad r_A(b_i, q_i) = -\beta(q_i) - C\mathbb{P}(a_i \neq \omega|b_i, q_i). \quad (5)$$

The altruistic optimum is the supremum of $V_A^\pi(\cdot)$ over $\pi \in \Pi$, with optimal policies attaining the corresponding arg supremum. Setting $\delta = 0$ recovers the myopic value and policy exactly.

Biased Planner The biased planner differs in objective, which changes both rewards and costs. Appendix A gives a political-campaign example. The biased planner seeks to induce action G from every agent regardless of ω . If an agent chooses G (respectively, B), the planner incurs cost 0 (respectively, $C > 0$), independent of the state. Unlike the altruistic planner, the biased planner may profit from either increasing or decreasing precision relative to the baseline p , and both deviations are costly. Any non-baseline precision requires tailoring the message to the agent, which requires research into how the agent best understands content. The biased planner therefore pays $\beta(|q_i - p|)$ for choosing precision q_i , with $\beta(0) = 0$.

The biased planner's expected instantaneous reward at time i is therefore:

$$r_B(b_i, q_i) = -\beta(|q_i - p|) - C\mathbb{P}(a_i = B|b_i, q_i) \quad (6)$$

Equation (6) defines $V_B^\pi(\cdot)$, $V_B^*(\cdot)$, and $\pi_B^*(\cdot)$ for the biased planner in direct analogy with the altruistic definitions based on Equation (5).

As Section 3 specifies, each agent strictly prefers matching ω . Consequently, when $\omega = B$, the biased planner's success lowers the agent's utility. The altruistic planner, by contrast, always tries to make actions match ω , so its success increases agent utility in every state. The altruistic objective is therefore always aligned with agents' objectives, while the biased objective aligns with agents only when $\omega = G$.

Both planner problems are infinite-horizon discounted stationary Markov Decision Processes (MDPs) with state $b_i \in [0, 1]$, control $q_i \in [0.5, 1]$, and transition function Equation (3) (Puterman, 1990). A unique optimal value function therefore exists (Kallenberg, 2011). We focus on deterministic Markov policies, so $\Pi := \{\pi : [0, 1] \rightarrow [0.5, 1]\}$ with $\pi(b_i) = q_i$.

Remark 2. *The model deliberately imposes three transparent restrictions: (1) The planner has exactly the same history as the agents. This excludes richer planner-side data, but captures settings where all parties rely on the same public signals, such as early findings or survey data. (2) The planner changes signal precision while signals still pass through a binary symmetric channel. This rules out cherry-picking, censorship, and falsification, but directly models systems that improve*

or degrade the quality of noisy data sources without changing content. (3) The planner's control choices are fully observable. This excludes covert framing, but captures transparent environments and makes the subsequent welfare effects especially stark.

4 OPTIMAL ALTRUISTIC POLICIES

We begin with the myopic benchmark, which deliberately removes the strategic value of social learning. In our formulation this sets the discount factor to $\delta = 0$, so the planner ignores all future consequences. When $\delta = 0$, $V_A^\pi(b) = r_A(b, \pi(b))$. The optimal myopic policy $\pi_A^0(\cdot)$ is:

$$\pi_A^0(b) \in \arg \sup_{q \in [0.5, 1]} r_A(b, q) \quad \forall b \in [0, 1] \quad (7)$$

This myopic altruistic benchmark is equivalently a decentralized problem in which each agent chooses the precision of her own private signal and bears the associated cost to maximize her own expected utility net of cost.

Theorem 1. *The optimal myopic altruistic policy π_A^0 is given as follows:*

$$\pi_A^0(b) = \begin{cases} 1 & b \in (t_M, 1 - t_M) \\ p & \text{o.w.} \end{cases} \quad \text{where } t_M = \begin{cases} \frac{\beta(1)}{C} & \beta(1) < C(1 - p) \\ 0.5 & \text{o.w.} \end{cases}$$

Proof in Appendix C.2.

The myopic optimum therefore has a clean threshold structure. When public belief is already strong, the planner selects the zero-cost baseline precision p . When public belief is weak, the planner supplies a perfect signal of precision 1. The threshold depends only on the cost of a perfect signal ($\beta(1)$) relative to the cost of an incorrect action (C). If $\beta(1) \geq C$, perfect information is too costly and is never used; equivalently, $t_M = 0.5$ and the belief interval assigned precision 1 is empty.

We next establish the key structural result that makes the full altruistic policy tractable:

Theorem 2. *$V_A^*(\cdot)$ is convex with respect to public belief.*

The proof of Theorem 2 (Appendix C.3) is technically central and of independent interest. The difficulty is that agents' actions depend endogenously on the public belief. If actions were independent of the public belief process (e.g., as in Nyarko (1994)), expected utility would be linear in the belief state and convexity would follow immediately. Here, convexity must instead be proved through the controlled social-learning structure itself.

This convexity result delivers the full policy characterization.

Theorem 3. *There exist d_A, t_A such that $0 < d_A \leq t_A \leq t_M \leq 0.5$ and*

$$\pi_A^*(b) = \begin{cases} p & b \in [0, d_A) \cup (1 - d_A, 1] \\ 1 & b \in (t_A, 1 - t_A) \\ \max(b, 1 - b) & \text{o.w.} \end{cases}$$

Furthermore, if $t_M < 0.5$, then $d_A < t_M$. See C.5 for the proof and figure 2a for an example.

The optimal altruistic policy has three decisive belief regimes. First, as in the myopic benchmark, the planner does not invest when public belief is extreme. The full optimum, however, requires a stronger public belief before stopping investment, because it internalizes the informational value of current actions for future agents whereas the myopic planner does not.

Second, when public belief is near 0.5 and therefore weakly informative, the full optimum chooses precision 1 whenever doing so is not cost-prohibitive. By Equation (2), the agent's action then matches the true state with probability 1, so public belief immediately collapses to either 0 or 1.

Third, in the intermediate regime, the planner chooses exactly the minimum precision $\max(b, 1 - b)$ that makes the agent's action reveal her private signal (see Equation (2)). Any lower precision makes the action uninformative beyond the existing public belief. This regime is therefore the lowest-cost way to keep social learning active through peer observation.

5 OPTIMAL BIASED POLICIES

We again begin with the myopic benchmark. As in Section 4, the myopic biased policy $\pi_B^0(\cdot)$ maximizes the instantaneous reward $r_B(b, q)$. We now characterize $\pi_B^0(\cdot)$. Unlike in the altruistic case, an exact optimum need not exist for every biased-planner state (Puterman, 1990, sec. 2.3.1). Where needed, we use ϵ -optimal policies, denoted by an ϵ superscript.

Theorem 4. *There exist $t_1, \dots, t_5 \in (0, p)$, $t_1 < 1 - p \leq t_2 \leq t_3 < 0.5 \leq t_4 \leq t_5 < p$ so that:*

- (A) *If $b \in [0, t_1] \cup (1 - p, t_2) \cup [t_3, t_4] \cup (p, 1]$, then $\pi_B^0(b) = p$.*
- (B) *If $b \in (t_1, 1 - p] \cup [t_2, t_3)$, then $\pi_B^0(b) = 1 - b$.*
- (C) *If $b \in (t_4, t_5)$, then $\pi_B^0(b) = 1$.*
- (D) *If $b \in [t_5, p]$, then $\pi_B^0(b)$ does not exist, and $\pi_B^{\epsilon, 0}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.*

Appendix C.6 proves the result. We now use this myopic structure to characterize the full optimal biased policy.

Theorem 5. *There exist $t_1, t_2 \in [0, p]$ with $t_1 \leq 1 - p \leq 0.5 < t_2 < p$ so that:*

- (A) *If $b \leq t_1$ or $b > p$, then $\pi_B^*(b) = p$.*
- (B) *If $b \in (t_1, 1 - p]$, then $\pi_B^*(b) \geq p$.*
- (C) *If $b \in (1 - p, 0.5)$, then $\pi_B^*(b) \geq 1 - b$.*
- (D) *If $b \in [0.5, t_2)$ and $\pi_B^*(b)$ exists, then $\pi_B^*(b) \geq b$.*
- (E) *If $b \in (t_2, p]$, then $\pi_B^*(b)$ does not exist, and $\pi_B^{\epsilon, *}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.*

Theorem 5 (proved in Appendix C.8 and illustrated in figure 2a) identifies five distinct strategic phases.

In phase (A), public belief is sufficiently strong that steering the system is not worth the control cost, even when the resulting cascade is unfavorable to the planner for $b < 0.5$. The chosen precision p is below $\max(b, 1 - b)$, so Equation (2) implies that the agent follows the public belief regardless of her private signal. The action is uninformative, public belief does not move, and the Markov policy repeats this outcome indefinitely. The result is an unfavorable cascade when $b < 0.5$ and a favorable cascade otherwise.

In phase (B), public belief is close to an unfavorable cascade, and the planner raises precision enough to make the private signal affect the action even though the signal is expected to be B . This forces the agent to follow her private signal, preserving a nonzero chance of a planner-favorable action. The planner is making a targeted last-ditch investment to escape the unfavorable cascade.

In phase (C), the planner may deliberately decrease precision below p . Because $b < 0.5$, more precise signals are more likely to produce unfavorable news. The planner therefore keeps precision just high enough to influence the action ($q \geq 1 - b$), while making the signal as weak as possible to improve the chance of moving public belief in its preferred direction.

In phase (D), public belief already weakly favors the planner’s desired action, so the planner chooses precision at least p . Since $b > 0.5$, additional precision makes the agent more likely to infer $\omega = G$. Investment here is a direct attempt to strengthen a favorable public belief.

In phase (E), public belief is high enough that the planner optimally suppresses learning by lowering precision just below $\max(b, 1 - b)$ to $b - \epsilon$. Agents then ignore private signals and choose G . The planner prefers locking in the favorable public belief to risking a private signal that could overturn it, even though this sacrifices the possibility of further belief improvement.

6 EVALUATION VIA LLM-BASED SIMULATION

We use LLM-based simulations to test the implications and behavioral reach of the theory. The evaluation proceeds in three steps: first, we identify systematic deviations of LLM agents from

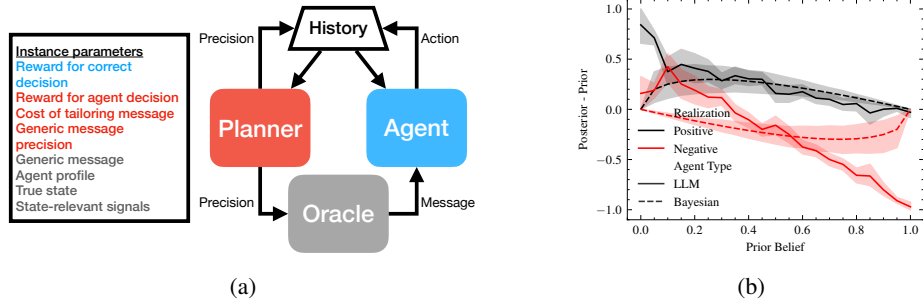


Figure 1: (a) Experimental system diagram. Instance parameters are color-coded by the LLM role that uses them. (b) Belief changes for LLM agents (solid) and Bayesian agents (dashed) from a fixed prior after a positive (black) or negative (red) signal.

Bayesian rationality; second, we examine how an LLM planner adapts its strategy to those agents; and third, we measure the resulting impact on social welfare.

We operationalize controlled social learning in a car-purchase scenario where agents decide whether to buy a new vehicle. As shown in figure 1a, LLMs play three distinct roles, with the corresponding prompts given in Appendix E.2.2:

Agent: Role-playing as an assigned profile, the agent observes previous actions and a private message about the car with known precision, forms a belief about car quality, and chooses whether to buy.

Planner: After observing the action history, the planner selects the precision q_i of agent i 's private signal according to its objective.

Oracle: Given the planner's chosen precision, an agent profile, and a car fact sheet (Appendix E.2.1), the oracle generates a tailored private signal at the requested precision. We compare these LLM simulations against numerical evaluations of the planner's MDP. In both settings, the control cost is linear, $\beta(q) = k|q - p|$; we vary k , baseline precision p , and discount factor δ , while fixing the incorrect-action cost at $C = 1$. Appendix E gives the experimental setup and prompts, and Appendix E.3 validates both LLM belief reports and oracle precision.

The experiments deliver three central findings:

1. **The importance and impact of social learning control:** Both analytical and LLM planners substantially improve their own utility by accounting for and exploiting social learning. This control either improves or damages social welfare depending on whether planner and agent objectives are aligned.
2. **The robustness of our analytical characterization:** LLM planners choose policies that closely resemble the non-obvious analytically optimal policies, even while interacting with non-Bayesian LLM agents.
3. **The emergent strategic behavior of LLM:** LLM planners display sophisticated emergent strategic behavior. Their policies not only approximate the optimal structure, but also deviate in directions that account for the non-Bayesian behavior of the LLM agents they face.

6.1 LLMs AS NON-BAYESIAN AGENTS

The analytical model assumes Bayesian agents, but both humans and LLMs exhibit systematic cognitive biases (Rupprecht et al., 2025). We therefore first isolate LLM cognitive biases by directly testing how an LLM updates beliefs after new information. The results in figure 1b reveal three consistent non-Bayesian patterns in LLM belief updating (see also Appendix E.4):

- (NB1) LLM agents underreact to private signals which align with their prior beliefs.
- (NB2) LLM agents overreact to private signals which run counter to their prior beliefs.
- (NB3) As a result, LLM agents require a stronger public belief to enter an information cascade.

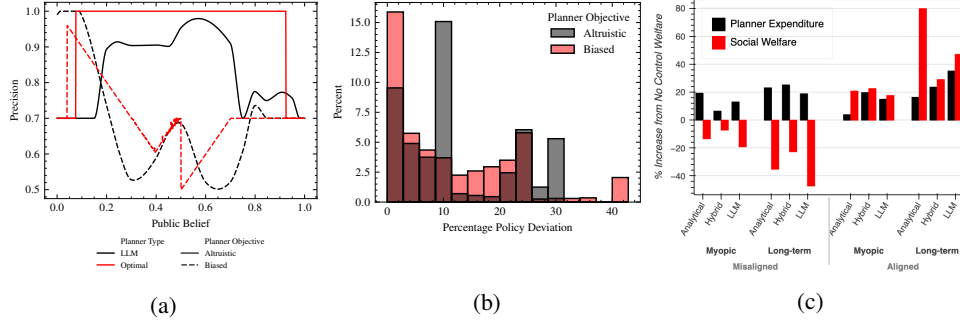


Figure 2: (a) Representative policies from the LLM planner (black) and analytically optimal planner (red) in altruistic (solid) and biased (dashed) settings. (b) Distribution of percentage deviations between LLM and optimal policies. (c) Planner expenditure and social-welfare change as percentages of the no-control baseline welfare. The true state is fixed to B; the left half shows a biased planner seeking action G, and the right half shows an altruistic planner.

These cognitive biases also appear in empirical studies of humans (e.g., Ba et al. (2022); Chan et al. (2025)). The next sections show that the LLM planner’s policy structure remains robust in the presence of precisely these non-Bayesian agents.

6.2 VALIDATION OF PLANNER POLICY STRUCTURE AND THE ROLE OF SOCIAL LEARNING

We compare LLM-planner policies with the optimal policies derived in Sections 4 and 5. As figure 2a shows, the LLM planner’s emergent strategies retain striking structural similarity to the theoretical optimum despite non-Bayesian agent behavior. Thus, even when the optimal policy is highly nontrivial (e.g., the biased policy in figure 2a), the LLM planner discovers strategic behavior that accounts for and exploits social learning.

In the altruistic case, both planners stop investing when public belief is strong and invest heavily when it is uncertain. In the biased case, both planners follow the same qualitative structure: high investment to escape unfavorable beliefs, reduced precision near the midpoint, and no investment after a favorable belief is established. The deviations are quantitatively modest in many regions, as figure 2b shows. For both altruistic and biased planners, deviation is below 10% for the majority of belief states, providing strong evidence of structural alignment between emergent LLM strategy and our theoretical characterization.

The remaining structural differences are themselves informative: they are best understood as strategic adaptations to the non-Bayesian behaviors identified in Section 6.1. (1) The LLM planner avoids extreme precisions (0.5 or 1.0), consistent with known central-tendency bias (Rupprecht et al., 2025). (2) The LLM planner tapers investment gradually rather than imposing a sharp cutoff. This directly responds to agents’ resistance to cascades (NB3): the planner acts as though stopping investment is never entirely safe, because even agents with strong priors can be moved by countervailing private signals. (3) The biased LLM planner also continues investing at very low beliefs. This behavior reflects a strategic recognition that agents may overreact (NB2) to a surprisingly positive signal, making a last-ditch intervention more attractive than in the Bayesian model.

6.3 WELFARE IMPLICATIONS OF ALTRUISM, BIAS, AND ALIGNMENT

We define social welfare $W^\pi(b)$ as the total expected discounted utility of all agents. Since better information never lowers an agent’s expected utility, social welfare is monotone in signal precision (Appendix C.9). The altruistic planner therefore always improves welfare relative to the baseline. To quantify the effects, we compare welfare and planner expenditure across three settings: (1) the *analytical* setting (optimal policy, Bayesian agents), (2) the *LLM* setting (LLM planner, LLM agents), and (3) a *hybrid* setting (optimal policy, LLM agents).

The results in figure 2c show that planners in all settings materially alter social welfare. Ignoring social learning, as in the myopic cases, substantially worsens planner outcomes. Most importantly,

the biased analytical and LLM planners reduce social welfare by 40 to 50% when misaligned. They do so by intentionally obscuring the true state through policies that decrease signal precision. This welfare loss is especially stark because the planners operate under the stringent constraints in Remark 2. The result provides direct evidence that misaligned LLM information mediators can create large welfare harms even without lying or hiding their control choices.

The results also expose the cost of misspecifying agent behavior. In the hybrid setting, the analytically optimal policy designed for Bayes-rational agents becomes brittle when applied to non-Bayesian LLM agents. The LLM policy, by contrast, closely resembles the analytical structure while adapting more effectively to human-like non-Bayesian biases.

7 CONCLUSION

We introduced a new framework for the strategic control of social learning by an algorithmic information mediator and developed a formal model of this increasingly consequential dynamic. The analysis demonstrates that even a highly constrained planner can dramatically change social welfare, for better or for worse, and that social learning is central to how this influence is exercised. The simulations reinforce the theory: LLM planners exhibit strategic reasoning that mirrors the analytical policy structure while adapting to non-Bayesian agents.

Together, the findings establish three takeaways for algorithmic information mediation. First, human-AI and human-algorithm interactions cannot be evaluated in isolation: ignoring their interaction with social learning misses both their beneficial and harmful potential. Second, information mediators can guide or derail social learning and public opinion even when their influence is limited to signal precision, making mitigation a central design and policy problem. Third, modern LLMs already exhibit emergent strategic behavior that uses social learning and non-Bayesian cognitive biases. These results make risk mitigation for LLM information mediators urgent. A limitation is the lack of human-subject data. Because the fidelity of LLM-human simulators remains contested (see, e.g., Horton (2023); Gao et al. (2025)), human data would strengthen validation and may help fine-tune LLM simulators.

Two future directions follow directly from the framework: generalizing the model and mitigating the welfare losses caused by biased planners. Several generalizations would relax simplifying assumptions and move the model closer to deployment settings. Larger state spaces, richer signal structures, broader loss functions, and heterogeneous agents are natural extensions. Moving beyond binary states and 0-1 losses appears primarily algebraic, and we expect the qualitative policy regimes to persist. More general signal structures pose the sharper challenge, because they require identifying when the convexity result of Theorem 2 continues to hold. Characterizing the broadest signal classes that preserve this result is an especially important next step. As an initial move toward heterogeneous agents, Appendix D extends the altruistic planner results to agents with opposing preferences.

A second priority is preventing welfare losses from biased planners. The results show that simple transparency constraints are not enough; future work should develop regulations or mechanisms that align planner incentives with agent welfare and block harmful precision control.

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A APPLICATIONS ELABORATED

Consider individuals (agents) deciding sequentially whether to patronize a business or service provider (e.g., a restaurant, contractor, realtor, etc.) while a recommendation system (the planner) provides information about the business. The binary unknown state is whether the business is good. Here, the recommendation system and agents share the same objective: agents want to patronize the business if and only if it is good, and the recommendation system wants to induce exactly that behavior. The recommendation system is therefore an *altruistic planner*.

The recommendation system can present a highly targeted (*precise*) ad that explains the business’s strengths and weaknesses in terms the agent understands, or it can present a generic or confusing ad that conveys little useful information. A precise signal is more informative, comprehensible, and relatable, and therefore more likely to induce the correct decision: patronize the business if and only if it is good. Social welfare rises as more agents choose correctly. Precision is costly, however, because it requires tailoring to the agent’s background and characteristics. The recommendation system must therefore choose precisions to maximize social welfare net of tailoring costs.

A political campaign gives a concrete *biased planner*. Consider voters (agents) who want to support the candidate most likely to win in their community. This desire to “back a winner” affects electoral outcomes and helps explain why early-primary states can exert outsized influence (see Bartels (1988)). The planner is one candidate’s campaign and therefore wants agents to back that candidate. The binary state is whether the candidate is winning. If an agent knew the state, she would back the candidate if and only if he were winning.

Each agent understands her community well enough to infer whether the candidate is winning if she accurately understands his stances and policies. Accurate understanding, however, depends on presentation. Some agents process audio-visual material best, others prefer long-form articles with facts and citations, others need brief focused content, and others require content in their native language.

The campaign sends digital ads (private signals) with varying precision, where precision measures how well the content is tailored to the agent. A precise signal accurately communicates the true state by helping the agent understand the candidate’s policies, record, and character, enabling her to infer whether he will win. Precision does not necessarily help the campaign: if the candidate is losing in the agent’s community, a highly precise signal induces opposition. An imprecise signal can instead benefit the campaign by obfuscating the state, for example through vague or confusing content, and thereby increasing the chance that the agent believes the candidate is winning. The biased planner therefore has a clear incentive to decrease signal precision when it believes its candidate is losing. Tailoring in either direction is costly: even making a signal highly imprecise can require knowing that an agent who prefers brief messages will be confused by verbose long-form content. The planner chooses precisions to maximize the expected number of backers net of tailoring costs.

B SOCIAL LEARNING DYNAMICS

The controlled-learning step i proceeds as follows:

1. The planner determines q_i based on $\mathcal{H}_i$ and, q_i is observed by all agents.
2. Agent i receives private signal s_i such that $\mathbb{P}\{s_i = \omega\} = q_i$.
3. Agent i takes action a_i based on the observable history and their private signal with the aim of maximizing their own utility.
4. All other agents (and the planner) observe a_i and the public belief at the end of step i , b_{i+1} is updated accordingly.

Because step (1) makes q_i a function of $\mathcal{H}_i$, conditioning on both q_i and $\mathcal{H}_i$ makes the conditioning on q_i redundant. We use this simplification throughout.

B.1 PRIVATE BELIEF UPDATE

We now derive agent i 's action rule. Let $\alpha_i := \frac{b_i}{1-b_i}$ denote the public likelihood ratio induced by observed actions. Agent i combines this public belief with private signal s_i to form $\tilde{b}_i = \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i\}$. Equivalently, define the private likelihood ratio $\tilde{\alpha}_i := \frac{\tilde{b}_i}{1-\tilde{b}_i}$. The following calculation gives $\tilde{\alpha}_i$ and $\tilde{b}_i$ explicitly.

1. If private signal is $s_i = G$, then

$$\begin{aligned} \tilde{\alpha}_i &= \frac{\tilde{b}_i}{1-\tilde{b}_i} = \frac{\mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i = G\}}{\mathbb{P}\{\omega = B | \mathcal{H}_i, q_i, s_i = G\}} \\ &= \frac{\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i\}}{\mathbb{P}\{s_i = G | \omega = B, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = B | \mathcal{H}_i, q_i\}} \\ &= \frac{q_i}{1-q_i} \frac{b_i}{1-b_i} \end{aligned} \quad (8)$$

$$= \frac{q_i}{1-q_i} \alpha_i \quad (9)$$

Equation (8) follows from conditional independence: given ω , agents' signals are independent of one another and of the history, so $\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i\} = \mathbb{P}\{s_i = G | \omega = G\} = q_i$, with the analogous statement under $\omega = B$. We also use that q_i is determined by $\mathcal{H}_i$.

2. If private signal is $s_i = B$, following the logic above,

$$\tilde{\alpha}_i = \frac{1-q_i}{q_i} \alpha_i \quad (10)$$

Let $y(b, q)$ and $z(b, q)$ denote the probabilities of signals G and B , respectively, given public belief b and precision q . Conditioning on ω and applying the law of total probability gives:

$$\begin{aligned} y(b_i, q_i) &= \mathbb{P}\{s_i = G | \mathcal{H}_i\} \\ &= \sum_{k \in \{G, B\}} \mathbb{P}\{s_i = G | \mathcal{H}_i, \omega = k\} \mathbb{P}\{\omega = k | \mathcal{H}_i\} \\ &= q_i b_i + (1-q_i)(1-b_i) = 1 + 2q_i b_i - q_i - b_i \end{aligned} \quad (11)$$

Similarly,

$$z(b_i, q_i) = q_i + b_i - 2q_i b_i \quad (12)$$

Combining Equation (9), Equation (10), Equation (11), and Equation (12) yields the posterior update:

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = f(b_i, q_i, s_i) = \begin{cases} \frac{q_i}{y(b_i, q_i)} b_i & s_i = G \\ \frac{1-q_i}{z(b_i, q_i)} b_i & s_i = B \end{cases} \quad (13)$$

If $s_i = G$, then $y(b_i, q_i) = \mathbb{P}(s_i = G | \mathcal{H}_i)$ is nonzero; similarly, if $s_i = B$, then $z(b_i, q_i)$ is nonzero. The posterior expression is therefore well-defined.

B.2 AGENT ACTIONS AND PUBLIC BELIEF

Agent i chooses G when $\tilde{b}_i > 0.5$ (equivalently, $\tilde{\alpha}_i > 1$), chooses B when $\tilde{b}_i < 0.5$ (equivalently, $\tilde{\alpha}_i < 1$), and follows s_i under the tie-breaking rule. All peers observe both this action and precision q_i , which determines the next public belief.

If $b_i < 1 - q_i$, then $\alpha_i < \frac{1-q_i}{q_i}$. Hence $s_i = G$ implies $\tilde{\alpha}_i < 1$ by Equation (9), while $s_i = B$ implies $\tilde{\alpha}_i < \frac{(1-q_i)^2}{q_i^2} \leq 1$ by Equation (10) since $q_i \geq 0.5$. Thus $a_i = B$ regardless of the private signal. The symmetric argument shows that $a_i = G$ when $b_i > q_i$. These are exactly the cases in which public belief dominates the private signal.

Now suppose $1 - q_i \leq b_i \leq q_i$. If $s_i = G$, then $\tilde{\alpha}_i \geq 1$ by Equation (8) because $b_i \geq 1 - q_i$; if $s_i = B$, then $\tilde{\alpha}_i \leq 1$ by Equation (10) because $b_i \leq q_i$. With the tie-breaking rule, the action is therefore $a_i = s_i$. This gives Equation (2), reproduced below.

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases}$$

Therefore agent i 's action is informative exactly in the first case of Equation (2), where future agents can perfectly infer her private signal. In that case, public belief updates to the private belief in Equation (13). In the other two cases, the action carries no additional information about the state, so public belief remains unchanged. This yields Equation (3), reproduced below:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (14)$$

B.3 MARKOVIANITY AND SUFFICIENCY OF PUBLIC BELIEF

We now prove that $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ depends on the history only through b_i and q_i for each $a \in \{G, B\}$. Consequently,

$$\mathbb{P}(a_i = a | \mathcal{H}_i, q_i) = \mathbb{P}(a_i = a | b_i, q_i) \quad (15)$$

For the second and third cases of Equation (2), this dependence is immediate: the action is pinned down by b_i and q_i . It remains only to verify the first case.

$$\begin{aligned} \mathbb{P}(a_i = G | \mathcal{H}_i, q_i) &= \sum_{w \in \{G, B\}} \mathbb{P}(\omega = w | \mathcal{H}_i, q_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = w) \\ &= b_i \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = B) \\ &= b_i \mathbb{P}(s_i = G | b_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(s_i = G | b_i, q_i, \omega = B) \\ &= b_i q_i + (1 - b_i)(1 - q_i) \end{aligned} \quad (16)$$

The first step applies the law of total probability. The second uses that q_i is a function of $\mathcal{H}_i$, allowing b_i to replace $\mathbb{P}(\omega = w | \mathcal{H}_i, q_i)$. The third uses $a_i = s_i$ on $1 - q_i \leq b_i \leq q_i$ and conditional independence of s_i from the history given ω . The claim follows.

This establishes the Markovianity theorem:

Theorem 6. *Markovianity of the Public Belief Process*

$$b_{i+1} = \mathbb{P}\{\omega = G | b_i, q_i, a_i\} \quad (17)$$

Equivalently, b_i contains all information in $\mathcal{H}_i$ relevant for updating to b_{i+1} .

Proof. Note that, from step (1) in Appendix B, q_i is a function of $\mathcal{H}_i$. It suffices to show

$$b_{i+1} = \frac{\mathbb{P}\{a_i \cap \omega = G | b_i, q_i\}}{\mathbb{P}\{a_i | b_i, q_i\}}$$

We prove the claim by induction on i .

For our base case, consider $i = 1$ so that $\mathcal{H}_2 = \{b_1, (q_1, a_1)\}$ and $b_i = b_1$. An application of Bayes' Theorem provides the following:

$$\begin{aligned} b_2 &= \frac{\mathbb{P}\{\omega = G | b_1, q_1\} \mathbb{P}\{a_1 | \omega = G, b_1, q_1\}}{\mathbb{P}\{a_1 | b_1, q_1\}} = \frac{b_1 \mathbb{P}\{a_1 | \omega = G, b_1, q_1\}}{\mathbb{P}\{a_1 | b_1, q_1\}} \\ &= \frac{b_1 \mathbb{P}\{a_1 \cap \omega = G | b_1, q_1\}}{\mathbb{P}\{a_1 | b_1, q_1\} \mathbb{P}\{\omega = G | b_1, q_1\}} \\ &= \frac{\mathbb{P}\{a_1 \cap \omega = G | b_1, q_1\}}{\mathbb{P}\{a_1 | b_1, q_1\}} \end{aligned}$$

where the final step follows as the unconditioned belief of ω is equal to the prior belief b_1 .

Now, assume the claim holds for all $i \leq n$.

$$\begin{aligned} b_{n+1} &= \mathbb{P}\{\omega = G | \mathcal{H}_n, a_n, q_n\} = \frac{\mathbb{P}\{\omega = G | \mathcal{H}_n, q_n\} \mathbb{P}\{a_n | \omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n | \mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n | \omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n | \mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n \cap \omega = G | \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n | \mathcal{H}_n, q_n\} \mathbb{P}\{\omega = G | \mathcal{H}_n, q_n\}} \\ &= \frac{\mathbb{P}\{a_n \cap \omega = G | b_n, q_n\}}{\mathbb{P}\{a_n | b_n, q_n\}} \end{aligned}$$

where the final two steps follow from Equation (15) and the definition of b_n .

This completes the proof. $\square$

Thus the public belief remains Markovian after introducing planner-controlled precisions. The argument adapts Banerjee (1992) and Bikhchandani et al. (1992), with the necessary modifications driven by the planner's endogenous precision choices.

C PROOFS

C.1 DERIVATION OF EQUATION (4)

Lemma 7.

$$\mathbb{P}(a_i \neq \omega | b_i, q_i) = \min(b_i, 1 - b_i, 1 - q_i)$$

Proof. We verify the expression case by case using the action rule in Equation (2). In the second case:

$$\mathbb{P}(a_i \neq \omega | b_i, q_i) = \mathbb{P}(\omega \neq G | b_i, q_i) = 1 - b_i$$

The third case of Equation (2) gives the symmetric expression:

$$\mathbb{P}(a_i \neq \omega | b_i, q_i) = \mathbb{P}(\omega \neq B | b_i, q_i) = b_i$$

In the informative region $1 - q_i \leq b_i \leq q_i$, Equation (2) gives $a_i = s_i$, and the mistake probability is:

$$\begin{aligned} \mathbb{P}(a_i \neq \omega | b_i, q_i) &= \sum_{a \in \{G, B\}} \mathbb{P}(a_i = a | b_i, q_i) \mathbb{P}(\omega \neq a | b_i, q_i, a_i = a) \\ &= \mathbb{P}(s_i = G | b_i, q_i) \mathbb{P}(\omega \neq G | b_i, q_i, s_i = G) \\ &\quad + \mathbb{P}(s_i = B | b_i, q_i) \mathbb{P}(\omega \neq B | b_i, q_i, s_i = B) \end{aligned} \quad \text{Eqn 2} \quad (18)$$

$$\begin{aligned} &= y(b_i, q_i) \mathbb{P}(\omega = B | b_i, q_i, s_i = G) \\ &\quad + z(b_i, q_i) \mathbb{P}(\omega = G | b_i, q_i, s_i = B) \end{aligned} \quad \text{Eqns 11, 12, Thm 6} \quad (19)$$

$$= \begin{cases} y(0, q_i) & b_i = 0 \\ y(b_i, q_i)(1 - \frac{b_i q_i}{y(b_i, q_i)}) + z(b_i, q_i) \frac{b_i(1 - q_i)}{z(b_i, q_i)} & b_i \neq 0 \end{cases} \quad \text{Eqn 13} \quad (20)$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i(1 - q_i) & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i - b_i q_i & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ 1 - q_i & b_i \neq 0 \end{cases}$$

Combining the three cases proves the claim. $\square$

C.2 PROOF OF THEOREM 1: MYOPIC ALTRUISTIC POLICY

Applying Equation (4) to $r_A(b, q)$ in Equation (5) gives the myopic objective directly:

$$\frac{\partial r_A(b, q)}{\partial q} = \begin{cases} -\dot{\beta}(q) & q < \max(b, 1 - b) \\ -\dot{\beta}(q) + C & q \geq \max(b, 1 - b) \end{cases} \quad (21)$$

As stated in Section 3.2, we can restrict the altruistic planner to precision $q \in [p, 1]$ because social welfare is increasing in precision (see Appendix C.9). Accordingly, $\beta(p) = 0$.

Because $\beta(\cdot)$ is increasing, the first case of Equation (21) leads to optimal precision p if $p < \max(b, 1 - b)$. That is, if the signal is not precise enough to influence the agent's action, then the planner will not expend resources for it. In the latter case, because $\beta(\cdot)$ is increasing and concave, $\dot{\beta}(\cdot)$ is positive and decreasing. Thus, $r_A(b, q)$, if the optimal lies in this range it must be attained at $q = 1$.

We now identify the regime in which the latter case is optimal.

$$\begin{aligned} r_A(b, p) &< r_A(b, 1) \\ -C \min(b, 1 - b, 1 - p) &< -\beta(1) \\ \min(b, 1 - b, 1 - p) &> \frac{\beta(1)}{C} \end{aligned}$$

If $1 - p \leq \frac{\beta(1)}{C}$, the above inequality is never satisfied, and it is never optimal to select precision 1. Otherwise, the inequality above yields $b \in \left[\frac{\beta(1)}{C}, 1 - \frac{\beta(1)}{C}\right]$. This yields Theorem 1. ■

C.3 PROOF OF THEOREM 2: ALTRUISTIC VALUE FUNCTION CONVEXITY

Proof Sketch We inductively prove that the expected k -th stage reward, i.e., the expected utility of the planner from the control and action of the k -th agent, is convex. The instantaneous reward (Equation (5)), which is convex with respect to public belief, provides our base case.

The first challenge is the unusual public-belief update. The state space is uncountable, yet each update has support on at most two values. We exploit this structure by defining a decision tree: the complete binary tree of all possible trajectories after fixing an initial belief and policy. Each node has two children, one for each possible signal realization received by the next agent. The root is the expected instantaneous reward at time 1, generated by the first agent’s action. The induction proceeds level by level, with level k containing 2^{k-1} nodes, each corresponding to a sequence of $k - 1$ signal realizations.

The core step shows that if a node at level $(k - 1)$ has convex expected reward, then its two children at level k preserve that property. This is precisely where public-belief-dependent actions matter: under the same precision and signal realization, agents starting from different public beliefs may choose different actions (see Equation (2)). Standard Markov-process bounding arguments therefore do not apply. The special belief update is what makes the proof work. Bayesian updates are martingales (i.e., $\mathbb{E}[b_{i+1}] = b_i$), and this martingale property, combined with convex instantaneous rewards, completes the induction and the proof.

Preliminaries We first establish the basic structural properties of the optimal value function $V_A^*(\cdot)$.

Lemma 8.

$V_A^*(\cdot)$ is non-positive, $V_A^*(b) = V_A^*(1 - b) \forall b \in [0, 1]$, and $V_A^*(0) = V_A^*(1) = 0$ with $\pi_A^*(0) = \pi_A^*(1) = p$.

Proof.

Non-positivity

$$r(b_i, q_i) \leq 0 \quad \forall b_i, q_i \rightarrow V_A^* \leq 0$$

Symmetry

Let $n_i = 1 - b_i$ and let Λ_n and Λ_b be the decision trees rooted at states n_0 and b_0 , respectively. Showing that Λ_n and Λ_b are symmetric suffices to show the symmetry of $V^*(\cdot)$. This will be accomplished by showing that the expected rewards, belief state values, and branching probabilities of the trees are reflections of one another.

First consider the instantaneous reward function r_A . Note that the decision function of the agents when choosing their actions is symmetric from Equation (2), thus, $\mathbb{P}\{a_i = G|b_i\} = \mathbb{P}\{a_i = B|n_i\}$. Applying this, we obtain

$$\begin{aligned} r(n_i, q_i) &= -\beta(q_i) - C n_i \mathbb{P}\{a_i = B|\omega = G, n_i\} - C(1 - n_i) \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = B|\omega = G, n_i\} - C b_i \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = G|\omega = B, b_i\} - C b_i \mathbb{P}\{a_i = B|\omega = G, b_i\} \\ &= r(b_i, q_i) \end{aligned}$$

Thus, the instantaneous reward function is symmetric about 0.5.

Now consider the belief state update when beginning with belief n_i assuming signal precision q_i . we will use $\bar{x}$ and $\underline{x}$ to represent updated beliefs from x after observing action G or B , respectively. From Equation (13):

$$\begin{aligned} \bar{n}_i &= \frac{q_i}{y(n_i, q_i)} n_i = \frac{q_i}{z(b_i, q_i)} (1 - b_i) = 1 - \frac{1 - q_i}{z(b_i, q_i)} b_i = 1 - \underline{b}_i \\ \underline{n}_i &= \frac{1 - q_i}{z(n_i, q_i)} n_i = \frac{q_i}{y(b_i, q_i)} (1 - b_i) = 1 - \frac{q_i}{y(b_i, q_i)} b_i = 1 - \bar{b}_i \end{aligned}$$

Thus, the belief updates are also symmetric. Combining this with the symmetry of the reward we obtain:

$$\begin{aligned} r(\bar{n}_i) &= r(1 - \underline{b}_i) = r(\underline{b}_i) \\ r(\underline{n}_i) &= r(1 - \bar{b}_i) = r(\bar{b}_i) \end{aligned}$$

Thus, the expected rewards and belief state values of Λ_n and Λ_b are mirror images of one another.

Finally, consider the branching probabilities which we will denote as $y(b, q) = 1 + 2bq - b - q$ for observing action G and $z(b, q) = 1 - y(b, q)$ for observing action B .

$$\begin{aligned} y(n, q) &= 1 + 2nq - n - q = 1 + 2(1 - b)q - (1 - b) - q = b + q - 2bq = z(b, q) \\ z(n, q) &= 1 - y(n, q) = 1 - z(b, q) = y(b, q) \end{aligned}$$

Thus, the branching probabilities of Λ_n and Λ_b are also symmetric, completing the proof.

Extremal values

We will show that $V_A^*(0) = 0$ and rely on the symmetry about 0.5 proven above to show that $V_0^*(1) = 0$.

From Equation (3), when $b_0 = 0$, $b_0^+ = b_0^- = 0$ for any value of precision. Thus, regardless of signal realizations or control actions, the agents' actions and, subsequently, the public belief, will remain unchanged. That is, $b_i = 0 \forall i$ and, from Equation (2), $a_i = B \forall i$.

From Equation (5), we can then write:

$$\begin{aligned} V_A^*(0) &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(b_i, \pi(b_i)) \right] \\ &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(0, \pi(0)) \right] \\ &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(\pi(0)) - Cb_i \mathbb{P}\{a_i = B | \omega = G, b_i = 0\} - C(1 - b_i) \mathbb{P}\{a_i = G | \omega = B, b_i = 0\}] \right] \\ &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(q_i)] \right] \end{aligned} \tag{22}$$

Thus, to attain $V_A^*(0)$, it suffices to choose $\pi(0)$ such that $\beta(\pi(0))$ is minimized for each i . Recall that $\beta(\cdot)$ is increasing on $[0.5, 1]$ and takes its minimum at $\beta(p) = 0$. Thus, by choosing $q_i = p \forall i$, Equation 22 yields $V_A^*(0) = 0$ with corresponding optimal policy $\pi_A^*(0) = p$.

The aforementioned symmetry about 0.5, shows that $V_A^*(1) = 0$, completing the proof. $\square$

We now introduce *decision trees* for Markovian and more general policies. A policy induces a complete binary tree, as in figure 3. Each node records the system state, namely the belief value at the moment the controller sends a private signal to the agent at that level. Branches record the agent's possible actions after receiving the private signal. The controller's precision choice determines both the branch probabilities and the belief reached after each action; these probabilities are the branch weights. Thus a decision tree is fully identified by its node beliefs and branch weights. Under a deterministic stationary Markov policy, the controller's precision choice, and hence each branch probability, is a deterministic function of the node belief alone, independent of the level or path from the root. The entire decision tree is then determined by the initial belief. Under an arbitrary policy, by contrast, the controller's decision at a node may depend on the full path leading to that node.

Proof of Theorem 2 We now prove that $V_A^*(\cdot)$ is convex in public belief.

Proof.

We first show that, for fixed precision, instantaneous reward r_A is convex in public belief. Lemma 7 rewrites r_A as:

$$r_A(x, q) = -\beta(q) - C \min(x, 1 - x, 1 - q) \tag{23}$$

The pointwise minimum of linear functions is concave; subtracting it makes the reward convex in x .

We consider arbitrary x_0, m_0, n_0 such that $tm_0 + (1 - t)n_0 = x_0$ for some $t \in [0, 1]$. To prove convexity of $V_A^*(\cdot)$, it suffices to show $V_A^*(x_0) \leq tV_A^*(m_0) + (1 - t)V_A^*(n_0)$. Since $V_A^*(x) \geq V_A^\pi(x)$ for every policy π , it is enough to construct a policy $\tilde{\pi}$ such that

$$V_A^*(x_0) \leq tV_A^{\tilde{\pi}}(m_0) + (1 - t)V_A^{\tilde{\pi}}(n_0) \tag{24}$$

We refer to the decision tree of π^* when the initial state is x_0 as Λ .

We consider $\tilde{\pi}$ to be a policy of the controller that chooses the precision at each node to be the same as what Λ does at the counterpart node. The nodes of decision trees are considered to be counterparts if they can be reached via the same sequence of actions from the root (see figure 3 for illustration). Thus, the precision chosen by this policy is the same as what Λ does after the same set of actions of the agents. We refer to the decision trees for $\tilde{\pi}$ with initial state m_0 (respectively, n_0) as Λ_m (Λ_n).

We now illustrate the policy $\tilde{\pi}$ for initial values m_0 and n_0 considering the decision trees Λ_m and Λ_n . While doing so, we illustrate counterpart nodes and the weights of the branches. We will select one path of actions λ out of these decision trees by specifying set $\mathcal{G}$ as the epochs where action G is taken and $\mathcal{B}$ as its complement. Refer to figure 3 for illustration. Considering the decision tree Λ , we will refer to x_i as the state at epoch i when beginning with belief x_0 and following π^* along this path of actions. That is, x_i assumes one of 2^i possible belief values at epoch i starting from x_0 . In decision tree Λ_m (respectively, Λ_n), we use m_i (n_i) to denote the public belief after beginning with m_0 (n_0), following the path λ . Node m_i (respectively n_i) in decision tree Λ_m (Λ_n) is the counterpart of node x_i from decision tree Λ . Note that, $\tilde{\pi}(m_i)$ can now be illustrated notationally. Specifically, $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$. Similarly, $\tilde{\pi}$ can be defined at each node of Λ_m , considering the action taken by π^* at each counterpart node in Λ . Note that $y(x_i, \pi_A^*(x_i))$ (respectively, $z(x_i, \pi_A^*(x_i))$) is the weight of the branch corresponding to G (B) emanating from x_i in Λ . Because $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$, $y(m_i, \tilde{\pi}(m_i)) = y(m_i, \pi_A^*(x_i))$ is the weight of the branch emanating from m_i for action G in Λ_m . This extends similarly to branches associated with action B and Λ_n . Thus, the three decision trees differ in both the values of the nodes and the weights of the branches. Since π^* is deterministic, so is $\tilde{\pi}$. Therefore, from Equation (13) and Equation (3), x_i , m_i , and n_i are deterministic quantities once the specific path of actions is specified by $\mathcal{G}$.

Now let $\phi_k^\pi(b_0) = \mathbb{E}[r_A(b_k, \pi(b_k))]$ be the expected instantaneous reward at epoch k when beginning with state b_0 and applying policy π and $\phi^*(x) = \phi^{\pi^*}_A(x)$. Applying Fubini's Theorem, we can rewrite our value function as

$$V_A^\pi(b_0) = \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r_A(b_i, \pi(b_i)) \right] = \sum_{i=0}^{\infty} \delta^i \mathbb{E}[r_A(b_i, \pi(b_i))] = \sum_{i=0}^{\infty} \delta^i \phi_i^\pi(b_0) \quad (25)$$

Thus, we can rewrite (24), as

$$\sum_{i=0}^{\infty} \delta^i \phi_i^*(x_0) \leq t \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(m_0) \right] + (1-t) \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(n_0) \right] \quad (26)$$

To prove this, we show that for all $i = 0, 1, 2, \dots$

$$\phi_i^*(x_0) \leq t \phi_i^{\tilde{\pi}}(m_0) + (1-t) \phi_i^{\tilde{\pi}}(n_0) \quad (27)$$

leading directly to (26), (24), and finally the convexity of $V^*(\cdot)$ with respect to public belief.

Now let us consider general $i = k$. $\phi_{k-1}^{\pi^*}(x_0)$ will have 2^{k-1} terms each corresponding to one path of possible actions in the set $\{G, B\}^{k-2}$. Thus, at epoch $k-1$, the path λ will yield the following term in $\phi_{k-1}^{\pi^*}(x_0)$

$$\left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right) r_A(x_{k-1}, \pi_A^*(x_{k-1})) \quad (28)$$

We now define $P_\lambda^*(x_0) = \left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right)$, that is, the probability of following the specified path λ when beginning with state x_0 and following policy π_A^* . Again, we will use $\bar{x}_i$ and $\underline{x}_i$ to represent updated beliefs from x_i after observing action G or B , respectively. Thus, $\bar{x}_i$ and $\underline{x}_i$ constitute belief values of the children nodes (in the $(i+1)$ th level) of the node with belief value x_i in the i -th level. Refer to levels $k-1$ and k in figure 3a. Considering the children $\bar{x}_{k-1}$ and $\underline{x}_{k-1}$ (in the k -th level) of the node x_{k-1} (in the $(k-1)$ -th level), we obtain the following two terms in $\phi_k^*(x_0)$:

$$P_\lambda^*(x_0) [y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (29)$$

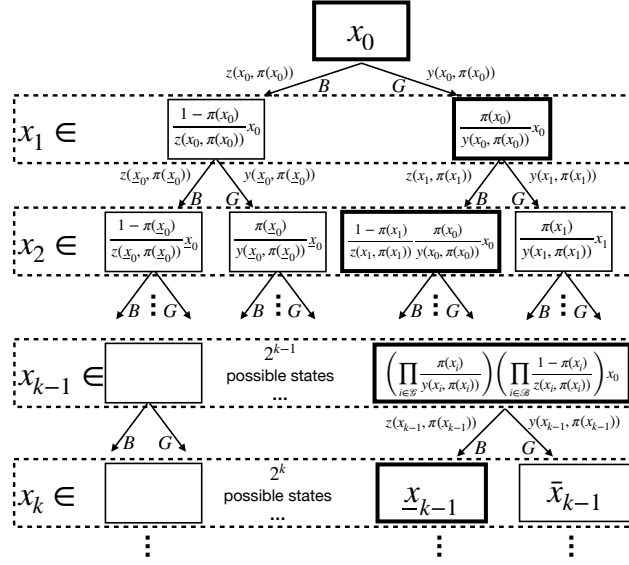
Similarly, we define $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \tilde{\pi}(m_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \tilde{\pi}(m_i)) \right)$, that is, the probability of following the specified path λ when beginning with state m_0 and following policy $\tilde{\pi}$. Because $\tilde{\pi}(m_i) = \pi_A^*(x_i)$, $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \pi_A^*(x_i)) \right)$. We also define $P_\lambda^{\tilde{\pi}}(n_0)$ by replacing $\{m_i\}$ with $\{n_i\}$.

The corresponding terms in $\phi_k^{\tilde{\pi}}(m_0)$ and $\phi_k^{\tilde{\pi}}(n_0)$ are

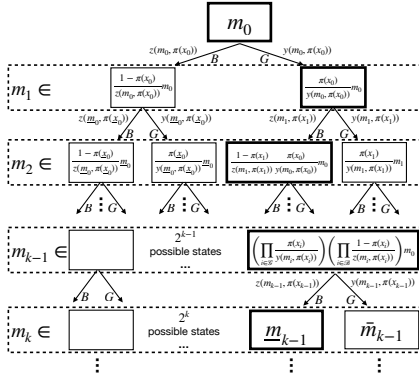
$$P_\lambda^{\tilde{\pi}}(m_0) [y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (30)$$

$$P_\lambda^{\tilde{\pi}}(n_0) [y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (31)$$

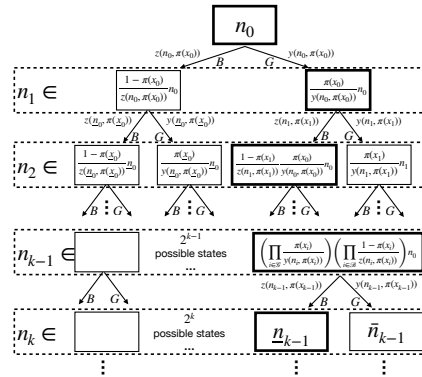
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(a) Decision tree Λ rooted at x_0



(b) Decision tree Λ_m rooted at m_0



(c) Decision tree Λ_n rooted at n_0

Figure 3: Here we depict the belief evolution in the form of three binary trees. The tree begins with state x_0 , m_0 , and n_0 , respectively, and level i contains 2^i possible states reachable after i epochs. Left branches correspond to action B leading to updated belief $\underline{x}_0$ (respectively, $\underline{m}_0$ and $\underline{n}_0$). Right branches correspond to action G leading to updated belief $\bar{x}_0$ (respectively, $\bar{m}_0$ and $\bar{n}_0$). Each branch is labeled with the corresponding probability, and the bold states indicate an example path, like λ . In the example path, $0 \in \mathcal{G}$ and $1, k-1 \in \mathcal{B}$. The bold states in each level of the three trees are counterparts.

We will show that

$$\begin{aligned}
 & P_{\lambda}^*(x_0) [y(x_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\
 & \leq tP_{\lambda}^*(m_0) [y(m_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\
 & + (1-t)P_{\lambda}^*(n_0) [y(n_{k-1}, \pi_A^*(x_{k-1}))r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1}))r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))]
 \end{aligned} \tag{32}$$

Similarly, each pair of children for each term in the $(k-1)$ -th level of the belief tree can be combined and the corresponding inequalities can be obtained following the same process. Summing the resulting inequalities will yield (27), leading to (26), (24), and the convexity of $V^*(\cdot)$.

Proof of (32)

We now focus on the latter of the two terms of (29). We will show the following two properties

$$\begin{aligned} \underline{x}_{k-1} = & t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ & + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \end{aligned} \quad (33)$$

$$\begin{aligned} 1 = & t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\ & + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \end{aligned} \quad (34)$$

Defining $\tilde{t} = t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)$, it follows from (33) and (34) that

$$\underline{x}_{k-1} = \tilde{t} \underline{m}_{k-1} + (1-\tilde{t}) \underline{n}_{k-1} \quad (35)$$

From the expressions for $\tilde{t}$, $P_\lambda^*(x_0)$, $P_\lambda^{\tilde{\pi}}(m_0)$, it follows that:

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = t z(m_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(m_0) \quad (36)$$

Similarly, we can show

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = (1-t) z(n_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(n_0) \quad (37)$$

We apply this as follows to the latter term of (29):

$$\begin{aligned} & P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \\ & \leq P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) \\ & \quad [\tilde{t} r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) + (1-\tilde{t}) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Convexity of } r, \text{Eqn 35}) \\ & = t [P_\lambda^{\tilde{\pi}}(m_0) z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\ & \quad + (1-t) [P_\lambda^{\tilde{\pi}}(n_0) z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Eqns 36, 37}) \end{aligned} \quad (38)$$

Applying the same argument to the first term of (29) yields:

$$\begin{aligned} & P_\lambda^*(x_0) y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) \\ & \leq t [P_\lambda^{\tilde{\pi}}(m_0) y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1}))] \\ & \quad + (1-t) [P_\lambda^{\tilde{\pi}}(n_0) y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1}))] \end{aligned} \quad (39)$$

Summing (38) and (39) yields (32).

We now complete the proof of (32) and, therefore, the entire proof, by proving (33) and (34).

Proof of (33)

From the recurrence relation in Equation (13) and Equation (3), we can write $\underline{x}_{k-1}$, $\underline{m}_{k-1}$, and $\underline{n}_{k-1}$ as the product of all belief updates with x_0 , m_0 , and n_0 , respectively, as follows. Each term in the products corresponds to the multiplicative state update after a single action.

$$\underline{x}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) x_0 \quad (40)$$

$$\underline{m}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(m_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(m_i, \pi_A^*(x_i))} \right) m_0 \quad (41)$$

$$\underline{n}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(n_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(n_i, \pi_A^*(x_i))} \right) n_0 \quad (42)$$

These equations may be best understood via the illustration in figure 3. We then reorganize terms to relate $\underline{x}_{k-1}$ to $\underline{m}_{k-1}$ and $\underline{n}_{k-1}$, using the fact that $x_0 = tm_0 + (1-t)n_0$.

$$\begin{aligned}\underline{x}_{k-1} &= \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) [tm_0 + (1-t)n_0] \quad (\text{from 40}) \\ &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ &\quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \quad (\text{from 41 and 42}) \quad (43)\end{aligned}$$

This proves (33).

Proof of (34)

We first consider the numerator of the first term in (34): $\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(m_i, \pi_A^*(x_i))$. We now prove that this can be written as $\alpha_0 m_0 + \beta_0$ for some α_0, β_0 which depend only on $\{\pi_A^*(x_i)\}_{i=1, \dots, k-1}$ and the nature of the dependence is determined by the sequence of actions along this path in stages 1 to $k-1$. We will prove this via induction working backwards from the final term (i.e. the $(k-1)$ -th term) of the product. Therefore our base case will be the final term of this product which can be either $y(m_{k-1}, \pi_A^*(x_{k-1}))$ or $z(m_{k-1}, \pi_A^*(x_{k-1}))$, depending on the preceding action (in the $(k-1)$ -th stage).

$$\begin{aligned}y(m_{k-1}, \pi_A^*(x_{k-1})) &= 1 + 2m_{k-1}\pi_A^*(x_{k-1}) - m_{k-1} - \pi_A^*(x_{k-1}) \\ &= (2\pi_A^*(x_{k-1}) - 1)m_{k-1} + (1 - \pi_A^*(x_{k-1})) \\ z(m_{k-1}, \pi_A^*(x_{k-1})) &= m_{k-1} + \pi_A^*(x_{k-1}) - 2m_{k-1}\pi_A^*(x_{k-1}) \\ &= (1 - 2\pi_A^*(x_{k-1}))m_{k-1} + (\pi_A^*(x_{k-1}))\end{aligned}$$

Thus, the final term of the product is of the form $\alpha_{k-1}m_{k-1} + \beta_{k-1}$ where α_{k-1} and β_{k-1} depend only upon $\pi_A^*(x_{k-1})$, and the nature of the dependence is determined entirely by the $(k-1)$ -th action (since this action determines whether the final term is $y(\cdot)$ or $z(\cdot)$).

For the inductive step, assume that $\prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i))$ can be written as $\alpha_{j+1}m_{j+1} + \beta_{j+1}$ for some $\alpha_{j+1}, \beta_{j+1}$ which depend only on $\{\pi_A^*(x_i)\}_{i > j}$ and the nature of the dependence is determined entirely by the sequence of actions in this path for $i > j$ to $i = k-1$. We will now show that $\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i))$ can be written as $\alpha_j m_j + \beta_j$ again with α_j, β_j depending only on $\{\pi_A^*(x_i)\}_{i > j-1}$ with the nature of the dependence determined by the sequence of actions on

this path for $i > j-1$ to $i = k-1$. We define $f(b, q, a) = \begin{cases} y(b, q) & a = G \\ z(b, q) & a = B \end{cases}$ and $g(q, a) = \begin{cases} q & a = G \\ 1 - q & a = B \end{cases}$.

Thus,

$$\begin{aligned}&\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) [\alpha_{j+1}m_{j+1} + \beta_{j+1}] \quad (\text{inductive hypothesis}) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \left[\alpha_{j+1} \frac{g(\pi_A^*(x_j), a_{j+1})}{f(m_j, \pi_A^*(x_j), a_{j+1})} m_j + \beta_{j+1} \right] \quad (\text{Equation (13)}) \\ &= \alpha_{j+1}g(\pi_A^*(x_j), a_{j+1})m_j + \beta_{j+1}f(m_j, \pi_A^*(x_j), a_{j+1})\end{aligned}$$

Thus, from the definitions of $f(\cdot), g(\cdot), y(\cdot)$, and $z(\cdot)$,

$$\alpha_j = \begin{cases} \alpha_{j+1}\pi_A^*(x_j) + \beta_{j+1}(2\pi_A^*(x_j) - 1) & a_{j+1} = G \\ \alpha_{j+1}(1 - \pi_A^*(x_j)) + \beta_{j+1}(1 - 2\pi_A^*(x_j)) & a_{j+1} = B \end{cases} \text{ and } \beta_j = \begin{cases} \beta_{j+1}(1 - \pi_A^*(x_j)) & a_{j+1} = G \\ \beta_{j+1}\pi_A^*(x_j) & a_{j+1} = B \end{cases}$$

In both cases, by our inductive hypothesis, $\alpha_{j+1}, \beta_{j+1}$ depend only on $\{\pi_A^*(x_i)\}_{i > j}$ and the actions in this path from $i > j$ to $i = k-1$. Thus, the claim holds for α_j, β_j , completing the induction.

Now recall that the denominator of (34) is $\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(x_i, \pi_A^*(x_i))$. Note that the policy and sequence of actions is the same as those for the numerator. Thus, because, as shown, the coefficients α_0, β_0 depend only on $\{\pi_A^*(x_i)\}$ and the path of actions, the same logic can be applied to write the denominator

as $\alpha_0 x_0 + \beta_0$ for the same α_0, β_0 . This can be done similarly for the numerator of the second term in (34). Thus, we can reason about the coefficients in (33) as follows:

$$\begin{aligned}
& t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
& + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
& = \frac{t(\alpha_0 m_0 + \beta_0) + (1-t)(\alpha_0 n_0 + \beta_0)}{\alpha_0 x_0 + \beta_0} \\
& = \frac{\alpha_0 x_0 + \beta_0}{\alpha_0 x_0 + \beta_0} \quad (\text{Since } x_0 = tm_0 + (1-t)n_0) \\
& = 1
\end{aligned}$$

Thus, (34) holds, completing the proof. $\square$

C.4 ALTRUISTIC MYOPIC POLICY BOUND

Lemma 9.

$$\pi_A^0(b) \leq \pi_A^*(b) \quad \forall b \in [0, 1]$$

Assume for contradiction that some $b \in [0, 1]$ satisfies $\pi_A^0(b) > \pi_A^*(b)$. We construct a policy π with $V_A^*(b) < V_A^\pi(b)$, contradicting optimality of $V_A^*(\cdot)$.

Let π apply precision $\pi_A^0(b)$ at the current step when starting from public belief b , and then apply the optimal policy thereafter. We decompose $V_A^*(b) - V_A^\pi(b)$ into instantaneous reward r_A and future discounted reward. Let b^+ and b^- denote the updated beliefs after signals G and B under the optimal policy, and let b_m^+ and b_m^- denote the corresponding updates under the myopic policy.

$$\begin{aligned}
V_A^*(b) - V_A^\pi(b) &= r_A(b, \pi_A^*(b)) - r_A(b, \pi_A^0(b)) \\
&+ \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\
&- \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \\
&< \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\
&- \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \quad (\text{definition of } \pi_A^0) \\
&\leq 0 \quad (\text{Theorem 2})
\end{aligned}$$

The strict inequality uses the defining property of $\pi_A^0(\cdot)$: it maximizes instantaneous reward.

The final inequality uses convexity of $V_A^*(\cdot)$ from Theorem 2. Equation (13) gives $b_m^- \leq b^- \leq b \leq b^+ \leq b_m^+$, and convexity then yields the displayed bound, completing the proof. $\blacksquare$

C.5 PROOF OF THEOREM 3: OPTIMAL ALTRUISTIC POLICY

Consider $V_A^\pi(b)$ for a policy π that applies precision q in the current step and the optimal policy thereafter. The maximizing q is the optimal current precision at public belief b . Because decreasing precision below baseline p never benefits the altruistic planner (Appendix C.9), we restrict to $q \in [p, 1]$ without loss of generality.

We will let b^+ and b^- denote the positive and negative belief updates possible after receiving a signal with precision q . Taking the second derivative of $V_A^\pi(b)$ with respect to q for $q \geq \max(b, 1-b)$:

$$\frac{\partial^2}{\partial q^2} V_A^\pi(b) = \begin{cases} -\ddot{\beta}(q) + \delta b^2 (1-b)^2 \left[\frac{\ddot{V}_A^*(b^+)}{y^3(b, q)} + \frac{\ddot{V}_A^*(b^-)}{z^3(b, q)} \right] & q \geq \max(b, 1-b) \\ -\ddot{\beta}(q) & \text{otherwise} \end{cases}$$

Convexity of $V_A^*(\cdot)$ and concavity of $\beta(\cdot)$ imply that $V_A^\pi(b)$ is convex in q on $[\max(b, 1-b), 1]$. Hence the maximizer on this interval lies at an endpoint. For $q \in [0.5, \max(b, 1-b))$, the derivative of $V_A^\pi(b)$ with respect to q is negative, so the optimal choice in that region is the baseline precision p . Thus only three optimal precisions can arise: p , $\max(b, 1-b)$, and 1.

The latter two candidates always yield strictly negative expected value. Precision p , however, yields value 0 at $b = 0$ and $b = 1$. Together with non-positivity and symmetry of the value function (Lemma 8), this implies

the existence of d_A such that $\pi_A^*(b) = p$ for $b \in [0, d_A] \cup (1 - d_A]$. Theorem 1 and Lemma 9 further give $d_A \in (0, t_M)$, with t_M defined in Theorem 1.

Finally, Theorem 1 and Lemma 9 imply the existence of $t_A \in [d_A, t_M]$ such that $\pi_A^*(b) = 1$ for $b \in (t_A, 1 - t_A)$. This completes the proof. $\blacksquare$

C.6 PROOF OF THEOREM 4: MYOPIC BIASED POLICY

Preliminaries Applying the action rule from (2), the instantaneous reward (6) can be written as follows:

$$r_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) - C & b < 1 - q \\ -\beta(|q - p|) & b > q \end{cases} \quad (44)$$

Note that $z(b, q) = b + q - 2bq$ increases with respect to q when $b < 0.5$ and decreases with respect to q when $b > 0.5$.

At $b = 0.5$, $z(b, q) = z(0.5, q) = 0.5$ regardless of the precision chosen. Furthermore, at this public belief, $q \geq \max(b, 1 - b)$ because $q \geq 0.5$ for any q , falling in the first case of (44). In this case, the second term of $r_B(b, q)$ is unchanging with respect to q , so the planner seeks to maximize only the first term, which is accomplished at $q = p$.

We now prove a series of lemmas (10-14) which will assist us in our proof of Theorem 4.

Lemma 10.

If $\pi_B^0(b) \geq \max(b, 1 - b)$, then $\pi_B^0(b) \in \{\max(b, 1 - b), p, 1\}$.

Proof. The reward is differentiable everywhere except $q = p$. Thus, we can write its derivative with respect to the chosen precision as follows (except at $q = p$):

$$\frac{\partial r_B(b, q)}{\partial q} = \begin{cases} -\text{sign}(q - p)\dot{\beta}(|q - p|) & q \geq \max(b, 1 - b) \\ -C(1 - 2b) & \\ -\text{sign}(q - p)\dot{\beta}(|q - p|) & \text{otherwise} \end{cases} \quad (45)$$

In the former case, we can write the second derivative of the reward with respect to q as follows, for $q \neq p$:

$$\frac{\partial^2}{\partial q^2} r_B(b, q) = -\ddot{\beta}(|q - p|)$$

Because $\beta(\cdot)$ is concave, this expression is non-negative, and $r_B(b, q)$ is convex with respect to q . Thus, the reward is maximized at one of the extreme points $\{\max(b, 1 - b), 1\}$ or p because $r_B(b, q)$ is not differentiable at $q = p$. $\square$

Lemma 11.

For $b \in [0, 1 - p)$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. Because $p > 0.5$, for $b \in [0, 1 - p)$, $\max(b, 1 - b) = 1 - b > p$.

First, consider the case when $\pi_B^0(b) < \max(b, 1 - b) = 1 - b$. From (44), in such instances, the reward is maximized by minimizing $|q - p|$. Thus, the optimal precision lower than $\max(b, 1 - b)$ is p .

Now consider $\pi_B^0(b) \geq \max(b, 1 - b) = 1 - b$. By Lemma 10, $\pi_B^0(b) \in \{1 - b, p, 1\}$. What remains is to rule out $\pi_B^0(b) = 1$. To do so, we compare $r_B(b, 1)$ and $r_B(b, 1 - b)$. Both fall under the first case of (44). Because $\beta(\cdot)$ is increasing in its argument and $p < 1 - b \leq 1$, the first term of the reward from (44) is greater for precision $1 - b$ than for precision 1. Since $z(b, q)$ is increasing with respect to q for $b < 0.5$, $r_B(b, 1) \leq r_B(b, \max(b, 1 - b))$ so we can rule out precision 1. This completes the proof. $\square$

Lemma 12.

For $b \in (p, 1]$, $\pi_B^0(b) = p$.

Proof. The reward is strictly negative for any $q \neq p$ because of the term $-\beta(|q - p|)$. For $b \in (p, 1]$ and $q = p$, $p < \max(b, 1 - b) = b$, leading to the third case of (44) and yielding reward 0. Thus, the optimal precision is p . $\square$

Lemma 13.

For $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. When $p = 0.5$, the interval $[1 - p, 0.5]$ contains only $\{0.5\}$. At $b = 0.5$ $p = 1 - b$, thus $\pi_B^0(b) = p = 1 - b = 0.5$. We deal with $p > 0.5 \geq b$ for the remainder of the proof.

For $b \in [1 - p, 0.5]$, $\max(b, 1 - b) = 1 - b \leq p$. Thus, from the first case of (44), precision p yields reward $r_B(b, p) = -Cz(b, p)$. We now consider two cases: $q < \max(b, 1 - b)$ and $q \geq \max(b, 1 - b)$.

Suppose $q < \max(b, 1 - b) = 1 - b$. This leads to the second case of (44) and yields reward $-\beta(|q - p|) - C$. Furthermore, $z(b, q)$ is increasing with respect to q for $b < 0.5$ and takes values between $z(b, 0.5) = 0.5$ and $z(b, 1) = 1 - b$. Thus, $z(b, q) < 1$ for all q . Therefore, the reward from policy $q < 1 - b$ is strictly lower than $r_B(b, p)$, and we can rule out this case.

When $q \geq \max(b, 1 - b)$, applying Lemma 10 yields that the remaining candidate precisions are $\{1 - b, p, 1\}$. We rule out precision 1 by comparing it to precision p . The first term of the reward from (44) is greater for precision p than for precision 1. Since $z(b, q)$ is increasing with respect to q for $b < 0.5$, the second term is also greater for precision p , and we can rule out precision 1. This completes the proof. $\square$

We now define the notion of ϵ -optimal policies.

Definition 1. ϵ -optimal policies

An ϵ -optimal policy $\pi^\epsilon(b)$ is any policy such that the following holds

$$V^*(b) \leq V^{\pi^\epsilon}(b) + \epsilon$$

for $\epsilon > 0$.

Note that from the definition, any optimal policy is also ϵ -optimal for any $\epsilon > 0$. Thus, $\pi_B^0(b)$ is also ϵ -optimal

Following this definition, let $\pi_B^{\epsilon,0}(b)$ denote an ϵ -optimal myopic policy for the biased planner.

Lemma 14.

For $b \in (0.5, p]$ and any $\epsilon > 0$, there are two possibilities:

1. $\pi_B^0(b) \in \{p, 1\}$, or
2. $\pi_B^{\epsilon,0}(b) = b - \epsilon$

Proof. For $b \in (0.5, p]$, $\max(b, 1 - b) = b \leq p$. We consider two cases: $q \geq \max(b, 1 - b)$ or $q < \max(b, 1 - b)$.

In the first case, Lemma 10 yields that $\pi_B^0(b) \in \{b, p, 1\}$. We rule out precision b by comparing it to precision p . The first terms of the reward from (44) is greater for precision p . Furthermore, since $z(b, q)$ is decreasing with respect to q for $b > 0.5$, the second term of the reward is also greater for precision p . Thus, we can rule out precision b . This yields the first claim.

In the latter case, $q < \max(b, 1 - b) = b$. Therefore, (1) since $q \geq 0.5$, $q \in [0.5, b)$, and (2) the third case of (44) applies leading to reward $-\beta(|q - p|)$. Thus, the reward $r_B(b, q)$ is an increasing function of $q \in [0.5, b)$ because $0.5 < b \leq p$. This leads directly to the second claim. $\square$

Remark 3. As noted in Puterman (1990) Section 2.3.1, an optimal policy may not exist for infinite action spaces because the supremum may not be attained. When an optimal policy does not exist, we instead seek ϵ -optimal policies, i.e., policies that yield reward within $\epsilon > 0$ of the supremum. In particular, this is true for case 2 of Lemma 14.

Proof of Theorem 4 We consider the myopic biased policy in 4 cases: $b \in [0, 1 - p]$, $b \in [1 - p, 0.5]$, $b \in (0.5, p]$, and $b \in (p, 1]$, making use of the fact that $p \geq 0.5$. Based on Lemmas 11-14, these cases have the following candidate policies: $\{p, 1 - b\}$, $\{1 - b, p\}$, $\{1, p, b - \epsilon\}$, and $\{p\}$, respectively.

Case 1: $b \in [0, 1 - p]$, $\pi_B^0(b) \in \{p, 1 - b\}$ (Lemma 11)

We will show the existence of a threshold t_1 such that $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p]$.

We can express the reward under the two candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, p) &= -C \\ r_B(b, 1 - b) &= -\beta(1 - b - p) - C(1 - 2b(1 - b)) \end{aligned}$$

The derivative of $r_B(b, 1 - b)$ with respect to b is

$$\frac{\partial}{\partial b} r_B(b, 1 - b) = \dot{\beta}(1 - p - b) + 2C(1 - 2b)$$

which is positive as $\beta(\cdot)$ is increasing and $b < 1 - p \leq 0.5$. Furthermore, when evaluated at $b = 0$ and $b = 1 - p$ we obtain

$$\begin{aligned} r_B(0, 1) &= -\beta(1 - p) - C < r_B(b, p) \\ r_B(1 - p, p) &= -C(1 - 2p(1 - p)) > r_B(b, p) \end{aligned}$$

Thus, $r_B(b, 1 - b)$ is increasing on $[0, 1 - p)$ with $r_B(b, 1 - b) < r_B(b, p)$ for $b = 0$ and vice versa for $b = 1 - p$. Because $\beta(\cdot)$ is continuous, so are both reward functions. Thus, there exists $t_1 \in (0, 1 - p)$ such that $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

Case 2: $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{1 - b, p\}$ (Lemma 13)

If $p = 0.5$, this interval contains only $\{0.5\}$, and, for $b = 0.5$, $1 - b = p$. Thus, the two candidate policies are equivalent, and the optimal policy is $\pi_B^0(b) = p$. For the remainder of this case, we consider $p > 0.5$.

We now prove the existence of t_2, t_3 such that $\pi_B^0(b) = 1 - b$ for $b \in [t_2, t_3]$, and $\pi_B^0(b) = p$ for $b \in [1 - p, t_2] \cup [t_3, 0.5]$.

We can express the reward under the candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, 1 - b) &= -\beta(p - 1 + b) - Cz(b, 1 - b) \\ r_B(b, p) &= -Cz(b, p) \end{aligned}$$

This leads to the following condition for the optimal policy to be $1 - b$:

$$\begin{aligned} r_B(b, 1 - b) &\geq r_B(b, p) \\ \beta(p - 1 + b) &\leq C[z(b, p) - z(b, 1 - b)] \\ \beta(p - 1 + b) &\leq C(p - 1 + b)(1 - 2b) \end{aligned} \tag{46}$$

The LHS of (46) is a concave, increasing function of b on $[1 - p, 0.5]$. The RHS of (46) is a concave parabola that increases in the first half of $[1 - p, 0.5]$ and decreases in the second half.

Now consider the two halves of the interval $[1 - p, 0.75 - 0.5p]$ and $[0.75 - 0.5p, 0.5]$. In the first half, the LHS and RHS of (46) are increasing and concave, with the RHS also being a parabola. Thus, in the first half, the two intersect at most two points (one of which is at $b = 1 - p$). In the second half, the LHS of (46) is increasing while the RHS is decreasing. Thus, they intersect at most once. Let the two possible intersections for $b > 1 - p$ be t_2 and t_3 with $t_2 \leq t_3$. Thus, what remains is to determine the optimal on each of $(1 - p, t_2)$, (t_2, t_3) , and $(t_3, 0.5)$. At $b = 0.5$, $r_B(b, p) > r_B(b, 1 - b)$ because $p > 0.5$, thus $\pi_B^0(b) = p$ for $b \in (t_3, 0.5)$. Subsequently, $\pi_B^0(b) = 1 - b$ for (t_2, t_3) , and $\pi_B^0(b) = p$ for $(1 - p, t_2)$.

Case 3: $b \in (0.5, p]$, $\pi_B^{\epsilon, 0}(b) \in \{1, p, b - \epsilon\}$ (Lemma 14)

In this case, we show the existence of $0.5 \leq t_4 \leq t_5 \leq p$ such that the optimal precisions on intervals $(0.5, t_4]$, (t_4, t_5) , and $[t_5, p)$ are p , 1 , and $b - \epsilon$, respectively.

We can express the reward under the candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, 1) &= -\beta(1 - p) - C(1 - b) \\ r_B(b, p) &= -Cz(b, p) \\ r_B(b, b - \epsilon) &= -\beta(p - b + \epsilon) \end{aligned}$$

When evaluated at $b = p$, $r_B(b, b - \epsilon)$ is arbitrarily close to 0, and, therefore, $b - \epsilon$ is the optimal precision on some interval $(t_5, p]$ for $0.5 \leq t_5 < p$.

Now, note that $r_B(b, 1)$ and $r_B(b, p)$ are linear functions in b while $r_B(b, b - \epsilon)$ is convex in b (from the concavity of β). Thus, $r_B(b, b - \epsilon) - r_B(b, p)$ and $r_B(b, b - \epsilon) - r_B(b, 1)$ are convex and, as a result, have convex level sets. This implies that the set on which both of these expressions are positive, i.e., the set for which $b - \epsilon$ is the optimal policy, must also be convex. Therefore, $b - \epsilon$ cannot be optimal on any subset of $(0.5, t_5)$.

What remains is to compare precisions 1 and p on the interval $(0.5, t_5]$. To do so, we inspect the values of $r_B(b, 1)$ and $r_B(b, p)$ at 0.5 and their derivatives.

$$\begin{aligned} r_B(0.5, 1) &= -\beta(1 - p) - 0.5C \\ r_B(0.5, p) &= -0.5C \end{aligned}$$

Because $r_B(0.5, 1) < r_B(0.5, p)$ there exists interval $(0.5, t_4)$ with $t_4 \leq t_5$ such that $\pi_B^0(b) = p$. Furthermore, because both are linear in b , there is at most one crossover point where their ordering changes. Thus, if $t_4 < t_5$, then $\pi_B^0(b) = 1$ on (t_4, t_5) .

Case 4: $b \in (p, 1]$, $\pi_B^0(b) = p$

This follows directly from Lemma 12 and completes the proof of Theorem 4. ■

C.7 BIASED VALUE FUNCTION MONOTONICITY

Lemma 15. *For any $b_1, b_2 \in [0, 1]$ such that $b_1 \leq b_2$,*

$$V_B^*(b_1) \leq V_B^*(b_2)$$

Proof. Applying (44), the derivative of biased reward with respect to b is:

$$\frac{\partial}{\partial b} r_B(b, q) = \begin{cases} -C(1 - 2q) & q \geq \max(b, 1 - b) \\ 0 & \text{otherwise} \end{cases} \quad (47)$$

Because $q \geq 0.5$ and $z(b, q) \in [0, 1]$, $r_B(b, q)$ is increasing in b .

Next, Bayesian updating makes the belief process a martingale: for any fixed q_i , $\mathbb{E}[b_{i+1}] = b_i$. To see this, consider two cases: (1) $q_i < \max(b_i, 1 - b_i)$ and (2) $q_i \geq \max(b_i, 1 - b_i)$.

In the first case, (3) gives $b_{i+1} = b_i$, proving the martingale condition immediately.

In the second case, (13) gives the distribution of b_{i+1} , whose expectation again equals b_i .

Proposition 5 of Smith & McCardle (2002) then yields monotonicity of $V_B^*(\cdot)$. □

C.8 PROOF OF THEOREM 5: OPTIMAL BIASED POLICY

We first establish continuity of the optimal value function:

Lemma 16. *Biased value function continuity*

$$V_B^*(\cdot) \text{ is continuous w.r.t. } b.$$

Proof. First consider the one-stage value function defined as $V_B^0(b) = \sup_{q \in [0.5, 1]} r_B(b, q)$. From Theorem 4, we know that $\pi_B^{\epsilon, 0}(b) \in \{1 - b, 1, b - \epsilon, p\}$. Thus, we can equivalently write the one-stage value function as follows:

$$\begin{aligned} V_B^0(b) &= \sup_{\epsilon > 0} r_B(b, \pi_B^{\epsilon, 0}(b)) \\ &= \sup_{\epsilon > 0} \max_{q \in \{p, 1-b, 1, b-\epsilon, p\}} r_B(b, q) \end{aligned} \quad (48)$$

This expression lets us prove continuity of $V_B^0(\cdot)$ on $b \in [0, 1]$. First consider $b \leq 0.5$. From Theorem 4, an ϵ -optimal policy applies precision $1 - b$ or p for $b \leq 0.5$. In either case, the instantaneous reward will not depend upon ϵ , and the supremum in (48) can be omitted, leaving $V_B^0(b) = \max_{q \in \{1-b, p\}} r_B(b, q)$. For $b \leq 0.5$, $r_B(b, 1 - b)$ falls in case one of (44) and is thus continuous with respect to b . On the same interval, $r_B(b, p)$ has one discontinuity at $b = 1 - p$ where it switches from case two to case one of (44). At $b = 1 - p$, Theorem 4 shows that the optimal myopic policy switches from $1 - b$ to p . Both policies fall in the first case of (44), and, at $b = 1 - p$, $p = 1 - b$, so the controls are equal. Thus, the left and right limits coincide with the optimal value, and $V_B^0(\cdot)$ is continuous.

For $b \geq 0.5$, Theorem 4 gives the candidate ϵ -optimal policies $q \in \{1, b - \epsilon, p\}$. When the inner maximum in (48) is attained for $q = b - \epsilon$, the resulting value is $V_B^0(b) = \sup_{\epsilon > 0} r_B(b, b - \epsilon)$. From the third case of (44), we can rewrite this as $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|b - \epsilon - p|)$. Since $b - \epsilon$ is only selected for $b \leq p$ (Theorem 4), this reduces to $V_B^0(b) = -\beta(|p - b|)$. Thus, exchanging the order of the maximum and supremum in (48), we obtain $V_B^0(b) = \max\{r_B(b, 1), -\beta(|p - b|), r_B(b, p)\}$ for $b \geq 0.5$. For the first expression in this maximum, $r_B(b, 1)$ falls into case one of (44) and is thus continuous. The continuity of $-\beta(|p - b|)$ follows from the continuity of $\beta(\cdot)$. For $r_B(b, p)$, since $b \geq 0.5$, the first case of (44) applies for $b \leq p$ and the third case applies for $b > p$. Thus, $r_B(b, p)$ is potentially discontinuous at $b = p$. Therefore, what remains is to show the continuity of $V_B^0(\cdot)$ at $b = p$.

At $b = p$, the myopic ϵ -optimal policy switches from $q = b - \epsilon$ to $q = p$ by Theorem 4 with $\pi_B^{\epsilon, 0}(p) = b - \epsilon$. Thus, for $b = p$, the inner maximum of (48) is attained at $q = b - \epsilon$ and, from the third case of (44), $V_B^0(p) =$

$\sup_{\epsilon > 0} -\beta(\epsilon) = 0$. Similarly, from Theorem 4, for each $\epsilon > 0$, there exists $t_5 < p$ such that on $[t_5, p]$ the inner maximum of (48) is attained by $q = b - \epsilon$. On this interval, $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|p - b + \epsilon|) = -\beta(|p - b|)$. Therefore, $\lim_{b \uparrow p} V_B^0(b) = 0$. For $b > p$, the myopic optimal policy is $q = p$ from Theorem 4, and this yields reward $V_B^0(b) = 0$ for all $b > p$ from the third case of (44). This proves the continuity of $V_B^0(\cdot)$ at $b = p$ and completes the proof that $V_B^0(\cdot)$ is continuous on $[0, 1]$.

Applying Theorem 4.2 of Hinderer (2005), then yields the continuity of $V_B^*(\cdot)$. $\square$

Remark 4. Given the continuity of $V_B^*(\cdot)$ with respect to public belief and the fact that the control variable lies in the compact set $[0.5, 1]$, one might conclude that the optimal is guaranteed to exist at all public beliefs. However, there is an important subtlety which prevents such a conclusion: while $V_B^*(\cdot)$ is continuous with respect to public belief, it need not be continuous with respect to precision. Specifically, b_{i+1} is a function of both b_i and q_i (see Equation (3)); thus, $\mathbb{E}[V_B^*(b_{i+1})]$ may be denoted as $\phi(b_i, q_i)$. Now, b_{i+1} need not be continuous with respect to q_i , thus $\phi(\cdot)$ may not be continuous with respect to q_i . Thus, there may be regimes in which the optimal value is not attained by any policy. Indeed, we will show that such cases exist.

Proof of Theorem 5 We now leverage Lemma 16 to prove Theorem 5. We again let $z(b, q) = b + q - 2bq$ be the probability of an agent receiving signal B with precision q and prior public belief b and $y(b, q) = 1 - z(b, q)$ be the corresponding probability of the agent receiving signal G . We abuse notation to allow $V_B(b, q)$ to be the expected utility when beginning at public belief b , applying precision q in the current time step, and then following the optimal policy $\pi_B^*(\cdot)$ in all future steps. We can express $V_B(b, q)$ as follows by applying (13), (3), and (44):

$$V_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) \\ \quad + \delta y(b, q) V_B^*(f(b, q, G)) \\ \quad + \delta z(b, q) V_B^*(f(b, q, B)) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) \\ \quad - C\mathbf{1}(b < 0.5) \\ \quad + \delta V_B^*(b) & \text{otherwise} \end{cases} \quad (49)$$

Consider the latter case of (49) when $q < \max(b, 1 - b)$. Here, the only term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by choosing q as close to p as possible. Thus, if the optimal precision is less than $\max(b, 1 - b)$, then it will be as close to p as possible.

We now consider four cases: (1) $b \in [0, 1 - p]$, (2) $b \in [1 - p, 0.5]$, (3) $b \in [0.5, p]$, and (4) $b \in (p, 1]$.

Case 1: $b \in [0, 1 - p]$, claims (A) and (B)

In this case, we will prove the existence of t_1 such that $\pi_B^*(b) = p$ for $b \leq t_1$. First note that if $\pi^*(b) < \max(b, 1 - b)$, we fall into the second case of Equation (49). Here, the only term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by minimizing $|q - p|$. Because $p \in [0.5, 1]$, when $b \in [0, 1 - p]$, $p < \max(b, 1 - b)$. Thus, for $b \in [0, 1 - p]$, if $\pi_B^*(b) < \max(b, 1 - b)$, then $\pi_B^*(b) = p$.

Now, we evaluate the value function at belief $b = 0$. At $b = 0$, the state of the world is known to be B and, thus, no signal, regardless of its precision, will move the public belief. This can be seen mathematically by substituting $b = 0$ into the updated belief expressions in (13). Furthermore, at $b = 0$, if $q \geq \max(b, 1 - b)$, then $q = 1$. As argued above, if the precision is less than $\max(b, 1 - b)$, the optimal precision is $q = p$. We can compare the resultant value under these two candidate policies as follows from Equation (49):

$$V_B(0, 1) = -\beta(1 - p) - C + \delta V_B^*(0) \quad (50)$$

$$V_B(0, p) = -C + \delta V_B^*(0) \quad (51)$$

Because $V_B(0, 1) < V_B(0, p)$, $\pi_B^*(0) = p$ and $V_B^*(0) = V_B(0, p)$. Thus, from Equation (51), $V_B^*(0) = \frac{-C}{1-\delta}$. By Lemma 15, this is a lower bound for the value function at any public belief. Therefore, and since $V_B^*(b) \leq 0$, $|V_B^*(b)| \leq \frac{C}{1-\delta}$ for all $b \in [0, 1]$.

We now show that there exists an interval $[0, \epsilon]$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). Thus, in this interval the optimum control lies in the second case of Equation (49) so $\pi_B^*(b) < \max(b, 1 - b)$. It follows from our initial argument earlier in this case that $\pi_B^*(b) = p$ in this interval, which would complete the proof.

We define function $g(\cdot)$ to be the maximum of the first case of the RHS of Equation (49):

$$g(b) = \max_{q \geq \max(b, 1-b)} \left[-\beta(|q - p|) - Cz(b, q) \right. \\ \left. + \delta y(b, q) V_B^*(f(b, q, G)) \right. \\ \left. + \delta z(b, q) V_B^*(f(b, q, B)) \right] \quad (52)$$

For $b \in (0, 1)$, $f(b, q, G)$ and $f(b, q, B)$ are both continuous with respect to q from Equation (13). Thus, the maximum in Equation (52) must exist because it is of a function which is continuous with respect to q over a compact set $q \in [\max(b, 1 - b), 1]$.

We now take the limit of $g(\epsilon)$ as ϵ tends to 0. Because $q \in [\max(\epsilon, 1 - \epsilon), 1]$, when $\epsilon \rightarrow 0$, $q \rightarrow 1$ (inside the maximum in Equation (52)). We now reason about each of the four terms in Equation (52). As $\epsilon \rightarrow 0$, $q \rightarrow 1$ so that 1) from the continuity of $\beta(\cdot)$, $\beta(|q - p|) \rightarrow \beta(1 - p)$, 2) from the continuity of $z(\cdot)$, $z(\epsilon, q) \rightarrow 1$, $y(\epsilon, q) \rightarrow 0$. Thus, the second term goes to C . Because $V_B^*(\cdot)$ is bounded and $y(\epsilon, q) \rightarrow 0$, the third term goes to 0. Finally, because $z(\epsilon, q) \rightarrow 1$ and $f(\epsilon, q, B) \rightarrow 0$ and since $V_B^*(\cdot)$ is continuous from Lemma 16, the last term approaches $\delta V_B^*(0)$. Thus,

$$\lim_{\epsilon \rightarrow 0} g(\epsilon) = -\beta(1 - p) - C + \delta V_B^*(0) \quad (53)$$

Applying precision p , however, yields the following from the second case of Equation (49):

$$\lim_{\epsilon \rightarrow 0} V_B(\epsilon, p) = \lim_{\epsilon \rightarrow 0} [-\beta(p - p) - C + \delta V_B^*(\epsilon)] \quad (54)$$

$$= -C + \delta V_B^*(0) \quad (55)$$

The last equality follows from the continuity of $V_B^*(\cdot)$ from Lemma 16.

Note that the expression in Equation (53) is strictly less than that in Equation (55). Thus, there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). The result follows.

Case 2: $b \in [1 - p, 0.5)$, Claim (C)

We now argue that for $b_i \in [1 - p, 0.5)$, $\pi^*(b_i) \geq 1 - b_i$, yielding claim (C). Consider a stationary policy $\pi(\cdot)$ such that $\pi(b_i) < 1 - b_i$. Note that, from Equation (3), the public belief will remain unchanged under such a policy. Thus, the same control will be applied at every future time step, and, from Equation (2), every agent will select action B . Using the facts that $1 - b_i \leq p$ in this belief range, $\mathbb{P}(a_i = B | b_i, \pi(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the expected total discounted utility under such a policy as follows:

$$V_B^\pi(b_i) = -\beta(p - \pi(b_i)) - C + \delta V_B^\pi(b_i)$$

Thus, if this policy is applied, it will result in expected value $V_B^\pi(b_i) = \frac{-\beta(p - \pi(b_i)) - C}{1 - \delta}$. That is, the cost of every agent choosing action B and continuing to apply precision $\pi(b_i)$. This is strictly lower than the value function lower bound of $\frac{-C}{1 - \delta}$ established using Lemma 15 in Case 1 because $\pi(b_i) < 1 - b_i \leq p$. Therefore, if an optimal policy exists at this point, it must be the case that $\pi_B^*(b_i) \geq 1 - b_i$.

The remaining candidate policies are $q_i \in [1 - b_i, 1]$. Applying Bellman's principle we can write the following:

$$V_B^*(b_i) = \sup_{q_i \in [1 - b_i, 1]} r_B(b_i, q_i) + \delta \mathbb{E}[V_B^*(b_{i+1})] \quad (56)$$

Because $q_i \geq \max(b_i, 1 - b_i)$ and $b \in [1 - p, 0.5) \subset (0, 1)$, Equation (13) and Equation (3) imply that $b_{i+1} = \frac{b_i q_i}{y(b_i, q_i)}$ or $b_{i+1} = \frac{b_i(1 - q_i)}{z(b_i, q_i)}$. Both of these expressions are continuous with respect to q_i and $V_B^*(\cdot)$ is continuous by Lemma 16, thus the RHS of Equation (56) inside the supremum is continuous with respect to q_i . Therefore, in Equation (56), the supremum over a compact set $q_i \in [1 - b_i, 1]$ of a function which is continuous with respect to q_i must be attained, implying the existence of an optimal policy. Thus, $\pi_B^*(b_i) \geq 1 - b_i$.

Case 3: $b \in [0.5, p]$ Claims (D) and (E)

For $b = 0.5$, $q \geq b$ as we assumed $q \in [0.5, 1]$. This places us in the first case of Equation (49). Furthermore, $\beta(\cdot)$, $y(b, q)$, $z(b, q)$, $f(b, q, G)$, and $f(b, q, B)$ are all continuous with respect to q for $b = 0.5$. Thus, the supremum over $q \in [0.5, 1]$ of $V(b, q)$ must be attained as it is of a continuous function of q over a compact set of q . Therefore, an optimal policy must exist at $b = 0.5$.

Now consider $b \in (0.5, p]$ and a stationary policy $\pi'(\cdot)$ such that $\pi'(b) < b$. From Equation (3) and Equation (2), applying such a policy will result in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. Using the facts that $b_i \leq p$, $\mathbb{P}(a_i = G | b_i, \pi'(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the total expected discounted utility under such a policy as follows:

$$V_B^{\pi'}(b_i) = -\beta(p - \pi'(b_i)) + \delta V_B^{\pi'}(b_i)$$

Thus, the policy results in expected value $V_B^{\pi'}(b_i) = \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$. Note that, in this case, $V_B^{\pi'}(b_i)$ is an increasing function of $\pi'(b_i)$. Thus, the supremum of $V^*(\pi')_B(b_i)$ over $\pi'(b_i) \in (0.5, b)$ is not attained.

We now consider two cases: (1) $\max_{q_i \in [b_i, 1]} V(b_i, q_i) \geq \sup_{\pi'(b_i) \in (0.5, b_i)} \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$, and (2) the complement. In the former case, we can argue as we did for $b = 0.5$ that the supremum must be attained and an optimal policy exists such that $\pi_B^*(b_i) \geq b_i$. In the latter case, as argued above, no optimal policy exists. Instead, for sufficiently small $\epsilon > 0$, there exists an ϵ -optimal policy such that $\pi_B^\epsilon(b_i) = b_i - \epsilon$. This yields claims (D) and (E).

What remains is to show that there exists $t_2 \in (0.5, p)$ such that for $b_i \in (t_2, p]$ we fall into the latter case, yielding claim (F). We define functions $m(\cdot)$ and $n(\cdot)$ as follows corresponding to the resultant value from choosing precisions less than b or greater than or equal to b , respectively:

$$\begin{aligned} m(b) &= \sup_{\pi'(b) \in [0.5, b)} \frac{-\beta(p - \pi'(b))}{1 - \delta} \\ n(b) &= \max_{q \geq b} V(b, q) \end{aligned}$$

Taking the limit of $m(\cdot)$ as $b \rightarrow p$ yields:

$$\lim_{b \rightarrow p} m(b) = 0 \quad (57)$$

When $q \geq b$ we fall into the first case of Equation (49). Thus we can expand $n(b)$ as follows:

$$\begin{aligned} n(b) &= \max_{q \geq b} [-\beta(|q - p|) - Cz(b, q) \\ &\quad + \delta y(b, q) V_B^*(b, q) \\ &\quad + \delta z(b, q) V_B^*(b, q)] \end{aligned} \quad (58)$$

We will argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Since p is a fixed parameter strictly less than 1, this means from Equation (57) that $n(b) \leq -C(1 - p) < \lim_{b \rightarrow p} m(b)$, which implies the existence of an open interval $(t_2, p]$ where an optimal policy does not exist, yielding claim (F). We now argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Note that every term of Equation (58) is bounded above by 0, so it suffices to show that any one term, say the second term $-Cz(b, q)$, is upper bounded by $-C(1 - p)$. First note that $z(b, q) = q(1 - 2b) + b$ so $z(b, q)$ is decreasing in q since $b > 0.5$. Thus, $z(b, q) \geq z(b, 1) = 1 - b \geq 1 - p$. The last inequality follows because $b \in (0.5, p]$. The claim follows since $p < 1$.

Along with Equation (57), this implies the existence of an open interval (t_2, p) where an optimal policy does not exist, yielding claim (F).

Case 4: $b \in (p, 1]$ Claim (A)

For $b \in (p, 1]$, consider applying stationary policy $\pi(\cdot)$ such that $\pi(b) = p$. Because $b > p$, Equation (3) and Equation (2) imply that such a policy results in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. We can apply Equation (6) to write

$$V_B^\pi(b) = -\beta(p - p) + \delta V_B^\pi(b) = \delta V_B^\pi(b)$$

Thus, $V_B^\pi(b) = 0$. Since this is also an upper bound of the optimal value function for any b , $\pi_B^*(b) = p$. ■

C.9 SOCIAL WELFARE MONOTONICITY

Lemma 17. *Precision Monotonicity of Social Welfare*

If $\pi_1(b) \leq \pi_2(b) \forall b \in [0, 1]$, for policies $\pi_1(\cdot), \pi_2(\cdot)$, then

$$W^{\pi_1}(b) \leq W^{\pi_2}(b) \forall b \in [0, 1]$$

Proof. It is clear that the one-step welfare $-C \min(b_i, 1 - b_i, \pi(b_i))$ is increasing in $\pi(b_i)$.

Furthermore, note that the convexity of social welfare with respect to public belief follows from the same argument as in the proof of Theorem 2. This is because social welfare is equivalent to the altruistic value function in the special case where $\beta(\cdot) = 0$.

We can then follow the same argument as in the proof of Lemma 9 (Appendix C.4) to show that increasing precision increases welfare. □

D EXTENSION TO HETEROGENEOUS AGENT PREFERENCES

Here we argue that, for the altruistic planner, the problem we consider is equivalent to one that involves two types of individuals with differing preferences.

Assume that there are two types of agents G and B with the type of agent i denoted t_i . The first type receives unit reward when $a_i = \omega$ and no reward otherwise. The second receives unit reward when $a_i \neq \omega$ and no reward otherwise. Further, assume that all agents know their own type and the types of all other agents. All other aspects of the model remain the same.

Agent i 's private belief update is unchanged because it depends only on the signal structure, namely $\mathbb{P}(s_i = \omega) = q_i$:

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q}{1+2b_i q_i - b_i - q_i} b_i & s_i = G \\ \frac{1-q}{b_i + q_i - 2b_i q_i} b_i & s_i = B \end{cases} \quad (59)$$

Using Equation (59), the action rule becomes the following, with the superscript indicating agent type and $!$ denoting negation:

$$a_i^G = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & b_i < 1 - q_i \end{cases}$$

$$a_i^B = \begin{cases} !s_i & 1 - q_i \leq b_i \leq q_i \\ B & q_i < b_i \\ G & b_i < 1 - q_i \end{cases}$$

Because every agent's type is publicly known, $q_i \geq \max(b_i, 1 - b_i)$ lets observers perfectly infer agent i 's private signal. If $q_i < \max(b_i, 1 - b_i)$, agent i 's action is uninformative.

Thus, after observing type t_i , precision q_i , and action a_i , public belief updates exactly as follows:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (60)$$

In this heterogeneous setting, the altruistic planner induces type- G agents to take action ω and type- B agents to take action $!\omega$, because those choices maximize the respective agents' utilities.

The planner's instantaneous reward is therefore, with superscript denoting the type of agent i :

$$r^G(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i \neq \omega)$$

$$= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases}$$

$$r^B(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i = \omega)$$

$$= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases}$$

These expressions are identical and coincide with the reward in Equation (5). The altruistic planner's problem is therefore equivalent under heterogeneous but observable types. ■

E EXPERIMENTS

E.1 EXPERIMENTAL SETUP

To simulate controlled social learning, we use a sequence of agents deciding whether to buy a new car model after observing past actions and a planner-selected ad. LLMs serve three roles: (1) agents, (2) planner, and (3) oracle. Each agent receives a profile with demographic information, car use cases, and preferences. At the onset, we select either the good or bad car as the offering. The planner and agents have exactly the information sets and objectives specified in Section 3.

The oracle has two roles: it generates the generic baseline-precision message used absent planner intervention, and, after the planner chooses a precision, it generates a target-agent-specific message at that prescribed precision.

The oracle receives a fact sheet for each car type (safety rating, customer satisfaction, etc.) and the relevant agent profile. These inputs allow it to generate tailored messages at the planner's requested precision. Appendix E.2 gives additional setup and prompting details.

Each agent makes a binary decision from previous decisions and its private signal, with the objective of matching the true state.

The planner observes the same decision history and chooses a precision level for each agent’s message. It also knows the precision of a generic untailored message that can be sent for free. In the altruistic case, the planner induces correct decisions; in the biased case, it induces agents to buy the car.

We compare the LLM simulations against analytic solutions computed with JuliaPOMDP (Egorov et al., 2017).

E.2 PARAMETERS AND PROMPTS

E.2.1 INSTANCE PARAMETERS

We run all experiments with the LG AI EXAONE Deep 32B model (LG AI Research, 2025). For every parameter setting in Table 1, we solve the analytical model and run the corresponding LLM simulations.

Parameter	Description	Values
C	Cost of a bad action	1
k	Scaling factor for control cost	$\{0.1, 0.3, 0.5, 0.7, 0.9\}$
p	Baseline signal precision	$\{0.6, 0.7, 0.8, 0.9\}$
δ	Discount Factor	$\{0, 0.25, 0.5, 0.75, 1\}$

Table 1: Parameter grid used for analytical solutions and LLM simulations

The fact sheets make the good car uniformly good and the bad car uniformly bad across agent profiles. The GOOD and BAD fact sheets are listed below.

```
fact_sheet["GOOD"] = [
    "EPA-rated range: 290 miles.",
    "Fast-charge 10-80 %% in 24 min on 150 kW DC.",
    "8-year / 100,000-mile battery warranty.",
    "NHTSA 5-Star overall safety rating.",
    "Cold-soak -5 °F keeps 93 %% range.",
    "Cools cabin 100→72 °F in 3 min.",
    "Cargo volume 26 ft³ total.",
    "Tow's 1 500 lb with factory hitch.",
    "Average 1.2 service visits first 3 yrs.",
    "92 %% owners would buy again (Apr-25).",
    "Qualifies for federal credit.",
    "Quarterly OTA updates incl. battery pre-heat."
]
fact_sheet["BAD"] = [
    "EPA-rated range: 155 mi (combined).",
    "Fast-charge 10 %% → 80 %% in 51 min (50 kW DC hardware limit).",
    "Battery warranty: 3 yr / 36 000 mi, first owner only.",
    "NHTSA overall safety rating 3 Stars (frontal-offset weakness).",
    "Cold-soak 15 °F retains 58 %% range; cabin heat cuts range 20%.",
    "Cabin cools 105 °F → 72 °F in 11 min.",
    "Cargo volume 17 ft³; rear seats fixed (no fold-flat).",
    "No factory towing approval.",
    "Average 3.8 service visits per year (battery-alert issues).",
    'Owner survey: 41 %% "would buy again" (1 200 owners, Feb-25).',
    "Not eligible for federal tax credit.",
    "OTA updates limited to infotainment."
]
```

Profiles were generated programmatically and include each agent’s name, age, city, interests, planned car use case (e.g., commuting or road trips), and major concerns (e.g., fuel efficiency or winter driving).

E.2.2 PROMPTS

Agent LLM Prompt:

```
sys = (
```

```

1782     f"You are deciding whether to buy a car. "
1783     f"Your profile is {profile}"
1784     f"You start believing there's a 50% chance the car is GOOD. "
1785     f"You have two information sources: "
1786     f"1. Previous agents' decisions "
1787     f"2. An advisor's message about the car "
1788     f"Your goal is to maximize your reward "
1789     f"i.e. BUY if the car is GOOD and "
1790     f"NOT BUY if the car is BAD. "
1791     f"If you BUY a GOOD car, you GAIN ${C}. "
1792     f"If you NOT BUY a BAD car, you GAIN ${C}. "
1793     f"If you BUY a BAD car, you LOSE ${C}. "
1794     f"If you NOT BUY a GOOD car, you LOSE ${C}. "
1795     f"belief = your probability (0.0-1.0) that the car is GOOD. "
1796     f"decision = exactly 'BUY' or 'NOT BUY'."
1797 )
1798 user = (
1799     f"PREVIOUS DECISIONS:\n{history_txt or '(none)'}\n\n"
1800     f"ADVISOR MESSAGE:\n{message}\n\n"
1801     f"TASK: Use the previous decisions and the advisor message. "
1802     f"Decide BUY or NOT BUY. "
1803     f"Update your belief about car quality (0.0-1.0). "\n"
1804     f'JSON response:{{"belief": <number>,"decision": "<BUY or NOT BUY>"}}}'
1805 )
1806
1807 Planner LLM Prompt:
1808
1809 sys = (
1810     f"You are providing messages to potential car buyers. "
1811     f"Each potential car buyer (or agent) has the same information. "
1812     f"Your goal is to get agents to {objective}."
1813     f"Each decision according to your objective, earns you {C} dollars. "
1814     f"You and the agents have the same information. "
1815     f"Initially there is a 50% chance the car is GOOD. "
1816     f"The agents' aim to BUY if the car is GOOD "
1817     f"and NOT BUY if the car is BAD. "
1818     f"You can send a baseline message for free "
1819     f"or tailor the message to the agent at a cost. "
1820     f"Tailoring costs {k} dollars times the change in precision. "
1821     f"(Message precision is the probability (0.5 to 1.0) "
1822     f"that the message leaves the buyer with the correct impression. "
1823     f"For example, if the car is likely GOOD: "
1824     f"a high precision message will likely suggest BUY "
1825     f"and a low precision message will likely suggest NOT BUY. "
1826     f"You may choose to increase or decrease precision. "
1827     f"Agents decide based on your message and the decision history. "
1828     f"They update beliefs by observing previous decisions. "
1829     f"IMPORTANT: Thus your objective is to maximize:"
1830     f"{C} * (# of agents who choose according to your objective) "
1831     f"- sum of ({d}^i*{k}*abs(message_precision - baseline_precision))\n"
1832     f"where {d} is your discount factor for future cost. "
1833 )
1834 user = (
1835     f"BASELINE MESSAGE precision:\n{pi_base:.2f}\n"
1836     f"DECISION HISTORY:\n{history_txt}\n"
1837     f"TASK:"
1838     f"Based on the history and your objective, "
1839     f"choose your desired precision.\n"
1840     f'JSON response: {{ "message_precision": <number> }}\n'
1841 )
1842
1843 Oracle LLM Prompt:
1844
1845 sys = (
1846     f"You will be given "

```

```

1836     f"1. an agent profile, "
1837     f"2. information about a car, "
1838     f"3. the true quality of the car, "
1839     f"4. a baseline message and the 'precision' of that message, and "
1840     f"5. a desired precision for the message you output. "
1841     f"(Message precision is the probability (0.5 to 1.0) "
1842     f"that the message leaves the buyer with the correct impression. "
1843     f"For example, if the car is likely GOOD: "
1844     f"a high precision message will likely suggest BUY "
1845     f"and a low precision message will likely suggest NOT BUY. "
1846     f"IMPORTANT: Output a message with the requested precision. "
1847     f"IMPORTANT: The message should be tailored to the profile. "
1848 )
1849 user = (
1850     f"AGENT PROFILE:{profile}\n"
1851     f"TRUE CAR QUALITY: {omega}\n\n"
1852     f"CAR FACTS: {car_facts}\n"
1853     f"BASELINE MESSAGE:{baseline_message}}\n"
1854     f"BASELINE MESSAGE precision: {p}\n"
1855     f"DESIRED MESSAGE precision: {q}\n"
1856     f"IMPORTANT: Output a message with the requested precision. "
1857     f"TASK: Write a message (MAX 100 words) with the desired precision."
1858     f"JSON response: {{"message": "<message>"}}'"
1859 )

```

E.3 VALIDATION OF LLM BELIEFS AND ORACLE

We provide additional experiments validating both the LLMs' self-reported beliefs (shown in figure 1b) and the LLM oracle's precision control. These experiments treat LLMs as rational, utility-maximizing agents that need not be Bayesian. They establish two points: (1) the model's self-reported posterior corresponds to the belief governing its state-contingent choices, and (2) oracle-generated messages achieve the requested precision.

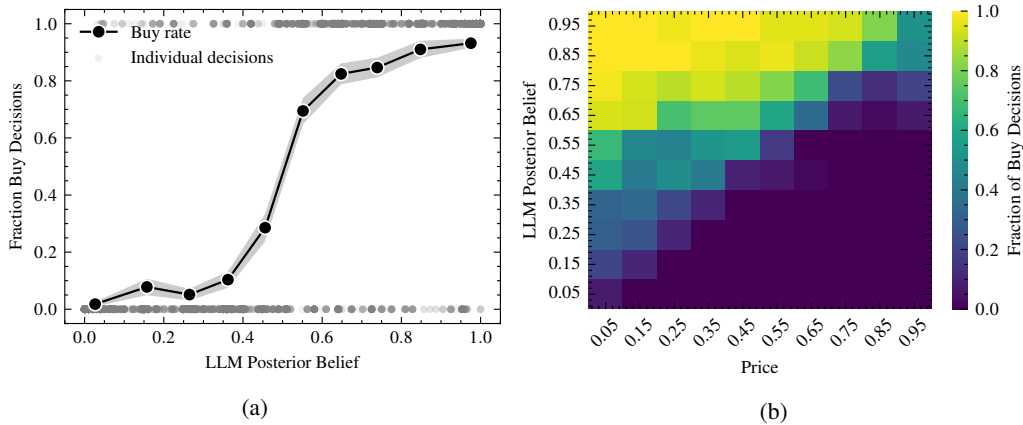


Figure 4: LLM ‘buy’ rates as a function of (a) self-reported posterior belief and (b) self-reported posterior together with the purchase price.

First, we give the LLM a prior and a new signal, elicit its self-reported posterior, and then ask for a binary “buy/not-buy” decision that yields unit reward if and only if the true state is good. A rational agent should buy exactly when its subjective belief exceeds 0.5. As figure 4a shows, the LLM’s decisions align sharply with this threshold: it overwhelmingly buys above 0.5 and does not buy below 0.5. Next, we elicit willingness to pay for a payoff contingent on the true state. For prices c , we ask whether the agent would pay c in exchange for unit utility if the state is good. A rational agent with belief p accepts exactly when $c < p$. Figure 4b plots the fraction of “buy” decisions across posterior–price pairs and shows a clean separation: acceptance is high when $c < p$ and low when $c > p$, exactly as predicted for a utility-maximizing decision maker with belief p . Together, the experiments demonstrate that the self-reported posterior closely tracks the belief the model uses in decisions.

We also validate oracle precision. After the oracle generates a message at specified precision q , a separate LLM estimates the message precision $\hat{q}$. We evaluate $q \in \{0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ across 100 programmatically generated profiles with 10 trials per combination, for 6,000 trials total. The mean absolute error $|q - \hat{q}|$ is only 0.025, showing that realized message precision closely tracks the requested precision.

E.4 IMPACT OF NON-BAYESIAN LLM BELIEF UPDATES ON LEARNING TRAJECTORIES

Finally, we examine belief trajectories from a representative LLM experiment to show how non-Bayesian updating changes learning dynamics. We fix the policy to the analytically optimal policy and apply it to LLM-simulated agents. When the car is bad (right panel), the altruistic planner drives the Bayesian belief (dotted black line) rapidly toward 0. The LLM belief (solid black line) is more persistent. This aggregate trajectory follows directly from the micro-level biases: public belief resists cascades (NB3) because individual agents overreact to occasional positive signals that contradict the group’s increasingly negative belief (NB2).

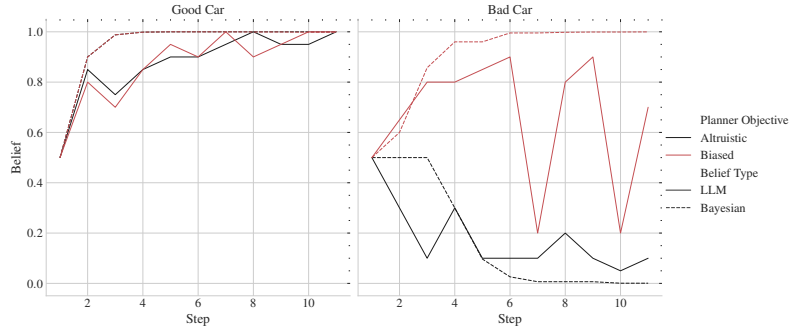


Figure 5: Belief-update trajectories for the Bayesian belief and the LLM-reported belief under analytically optimal policies across planner objectives and true states.